MBG 1201 Introduction to Modern Biology

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Outline

- DNA tools and biotechnology
- DNA sequencing
- Gene cloning
- PCR amplification
- Expressing genes
- Analyzing gene expression
- Gene editing

Terminology

- Genetic engineering: the direct manipulation of genes (insertion, mutation, deletion).
- **Nucleic acid hybridization**: the base pairing of one strand of a nucleic acid to a complementary sequence from another nucleic acid strand, either DNA or RNA. Nucleic acid hybridization forms the foundation of virtually every technique used in genetic engineering
- **Recombinant DNA**: Combination of nucleotide sequences from two different sources, often two species in vitro (in a test tube) into the same DNA molecule
- **Biotechnology:** the manipulation of organisms or their genetic components to make useful products

Applications of genetic engineering

Agriculture



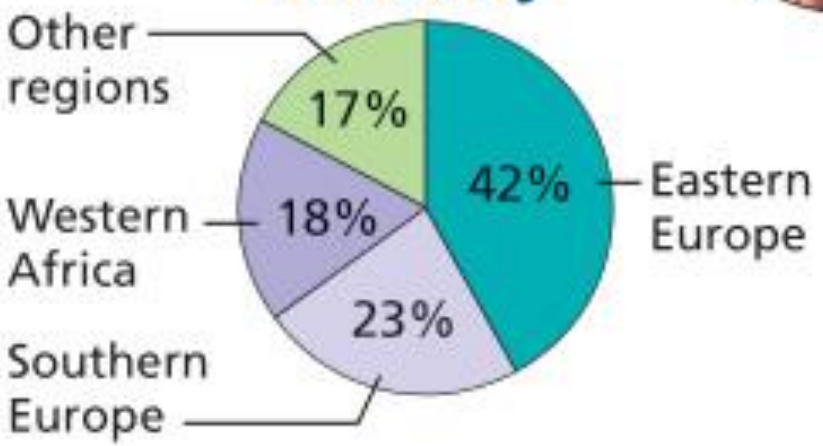
Medicine



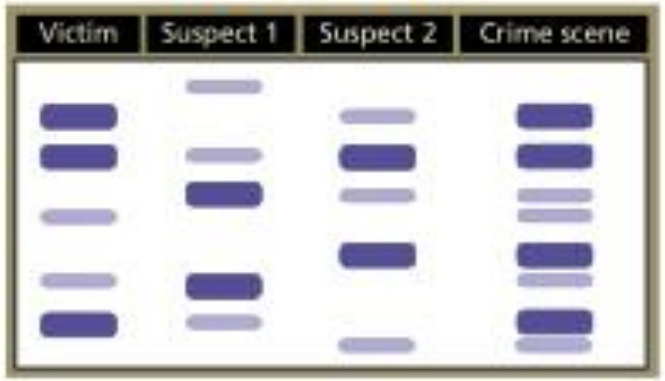
Environmental cleanup



Ancestry



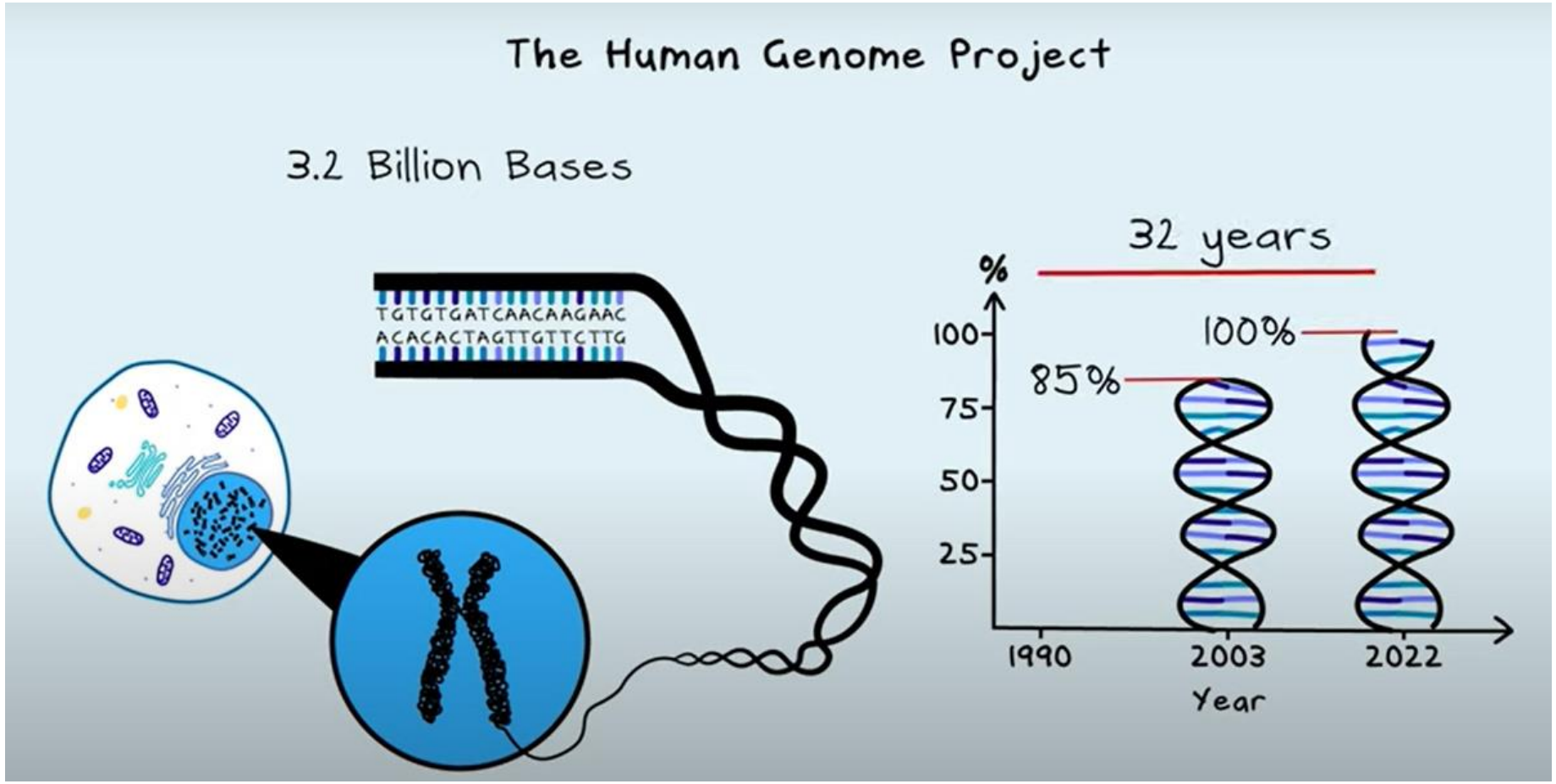
Forensics



Sequencing

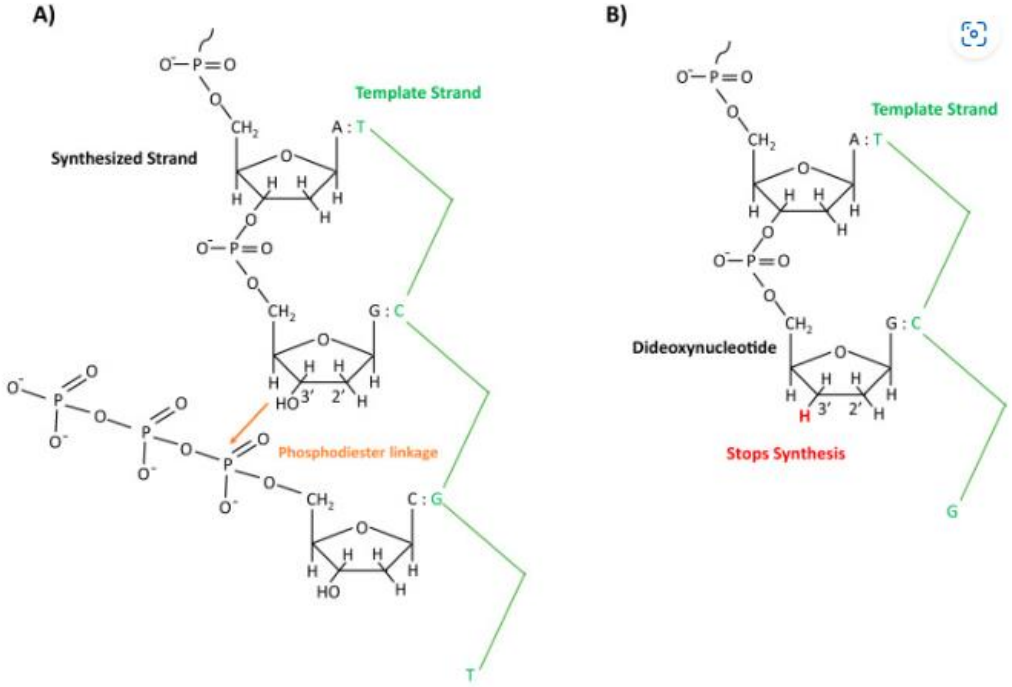
DNA sequencing is process to determine the complete nucleotide sequence of a DNA molecule.

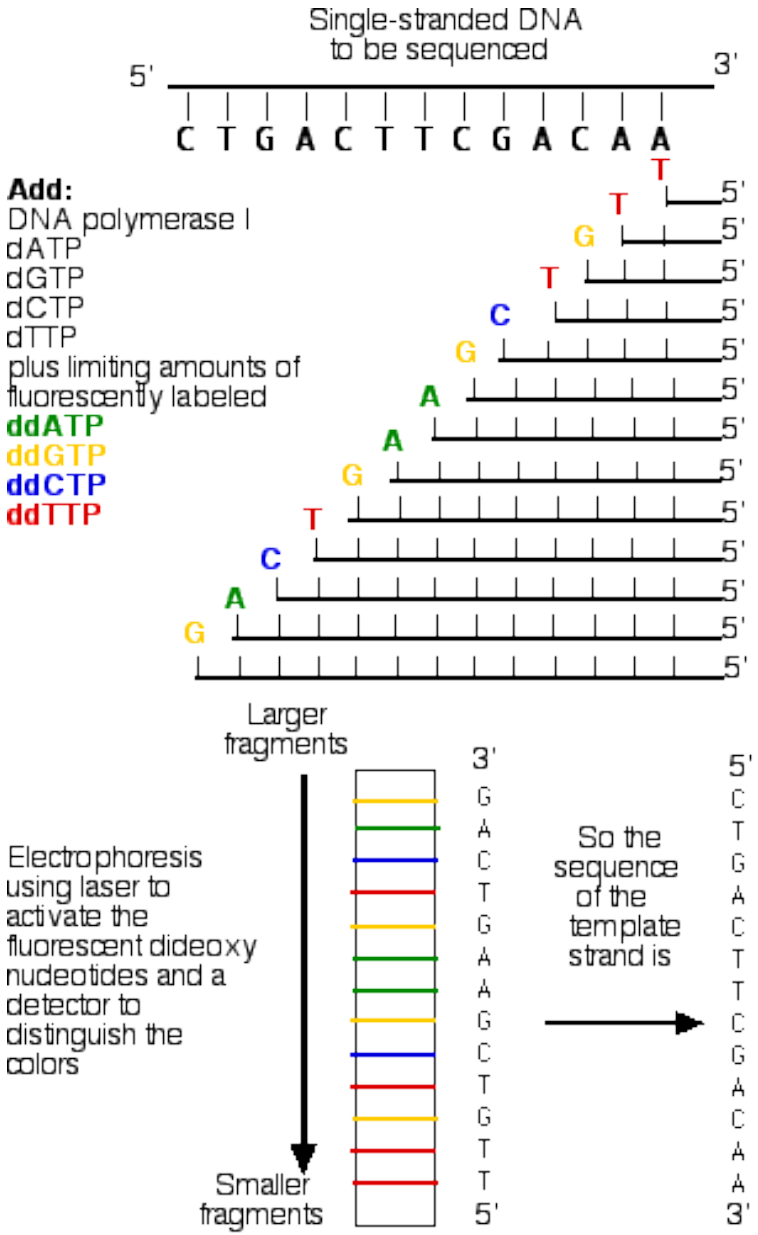
Sequencing of human genome is called the Human Genome Project. It started in 1990's with the contribution of scientist from around the world and took 32 years to complete.



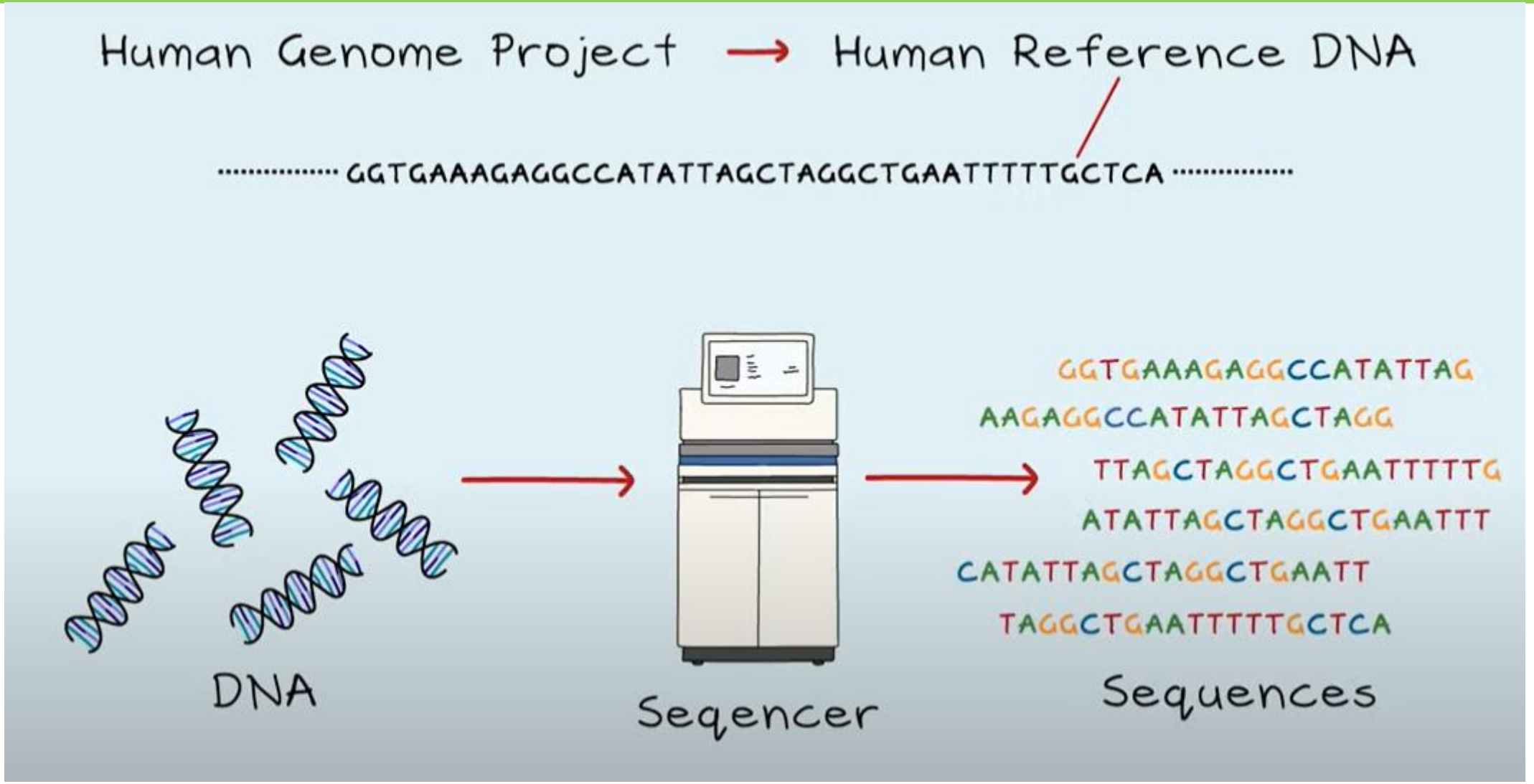
Sanger Sequencing

The method which allowed sequencing of the whole human genome is called Sanger sequencing (Dideoxy sequencing).
It was developed by Frederic Sanger in 1970s.
In Dideoxy nucleotides the OH groups in both 2’ and 3’ carbons are absent so DNA chain can not grow.



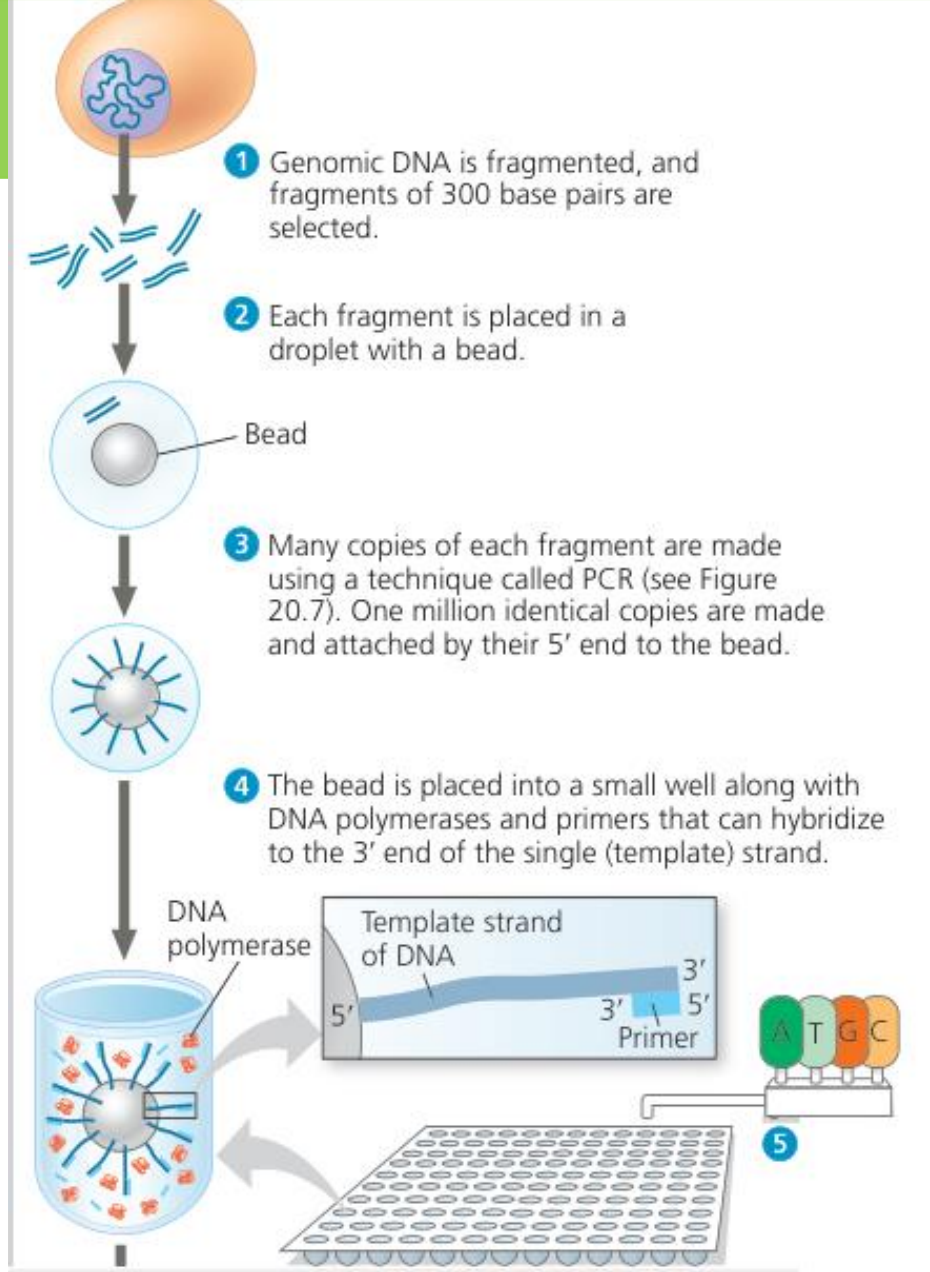


Next Generation Sequencing



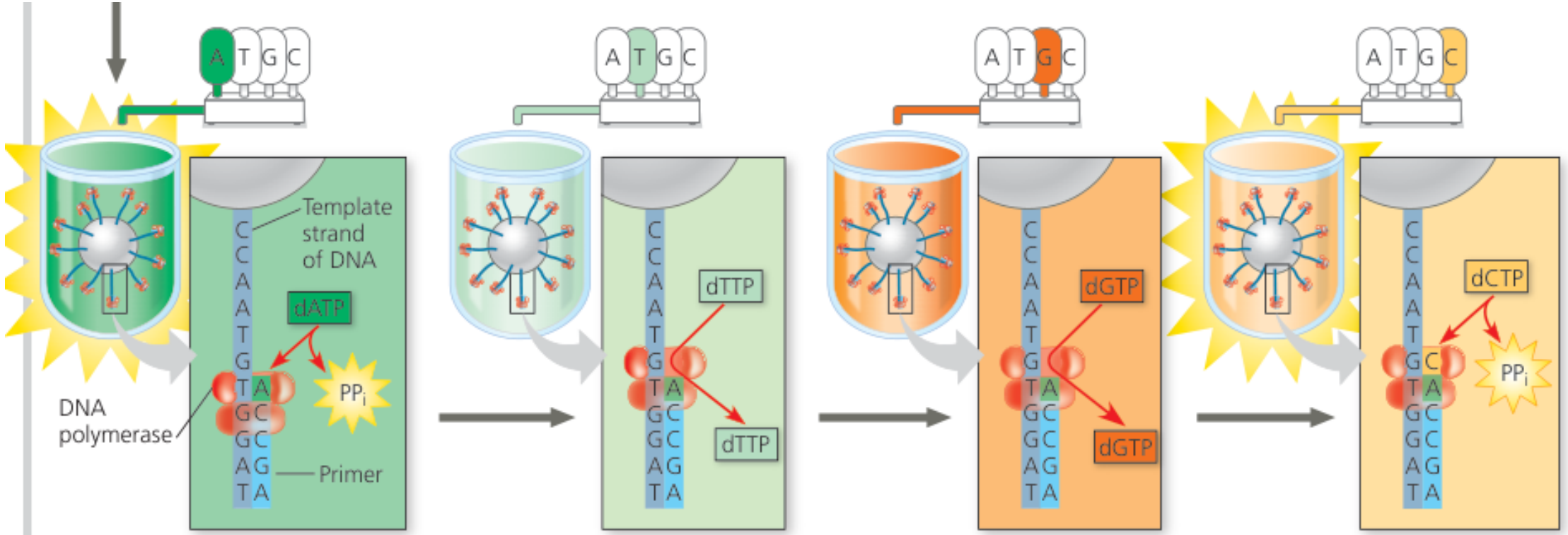
Whole genome is sequenced in one day.

Next Generation Sequencing: Sequencing by Synthesis



The well is one of 2 million on a multiwell plate, each containing a different DNA fragment to be sequenced. A solution of one of the four nucleotides required for DNA synthesis (deoxynucleoside triphosphates, or dNTPs) is added to all wells and then washed off. This is done sequentially for all four nucleotides: dATP, dTTP, dGTP, and then dCTP. The entire process is then repeated over and over again.

Next Generation Sequencing: Sequencing by Synthesis



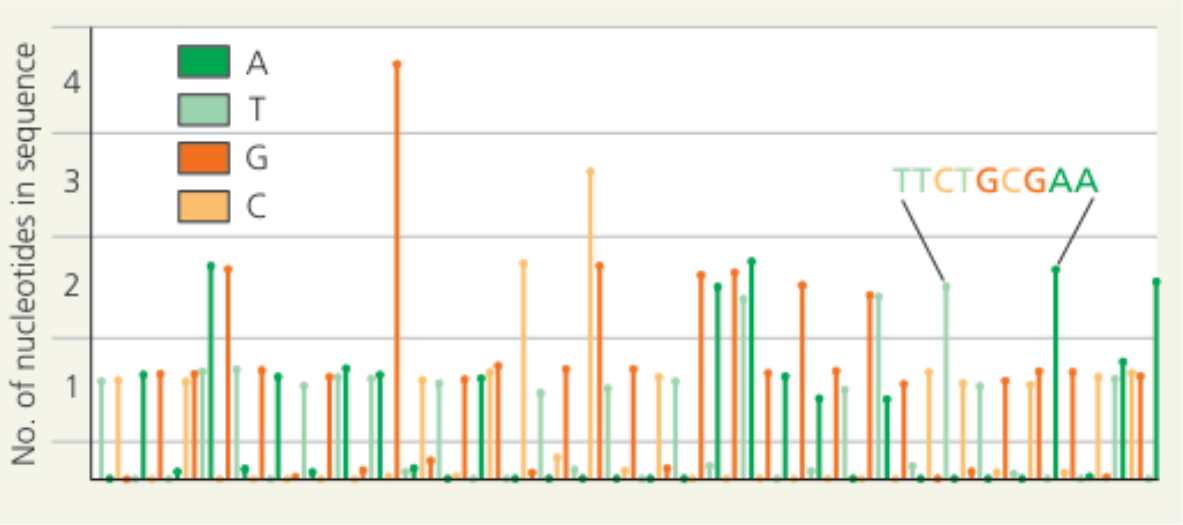
6 In each well, if the next base on the template strand (T in this example) is complementary to the added nucleotide (A, here), the nucleotide is joined to the growing strand, releasing PP_i, which causes a flash of light that is recorded.

7 The nucleotide is washed off and a different nucleotide (dTTP, here) is added. If the nucleotide is not complementary to the next template base (G, here), it is not joined to the strand, no reaction occurs, and there is no flash.

8 The process of adding and washing off the four nucleotides is repeated until every fragment has a complete complementary strand. The pattern of flashes reveals the sequence of the original fragment in each well.

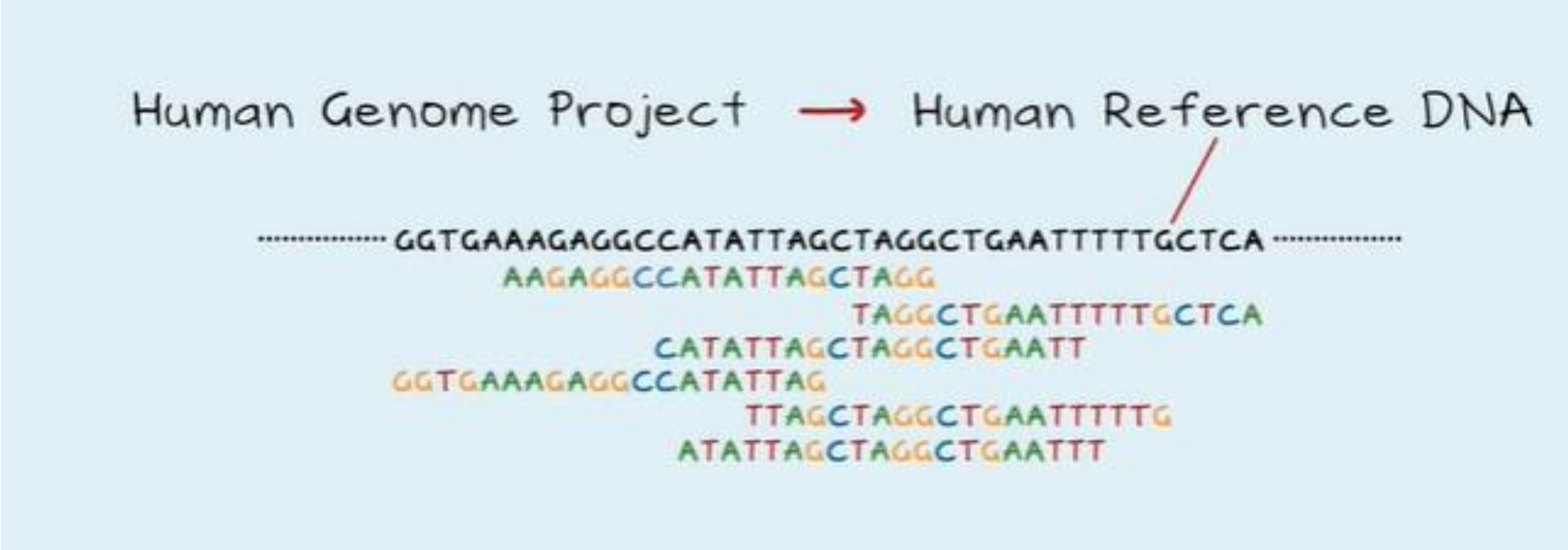
Next Generation Sequencing: Sequencing by Synthesis

Results Each of the 2,000,000 wells in the multiwell plate, which holds a different fragment, yields a different sequence. The results for one fragment are shown below as a "flow-gram." The sequences of the entire set of fragments are analyzed using computer software, which "stitches" them together into a whole sequence—here, an entire genome.



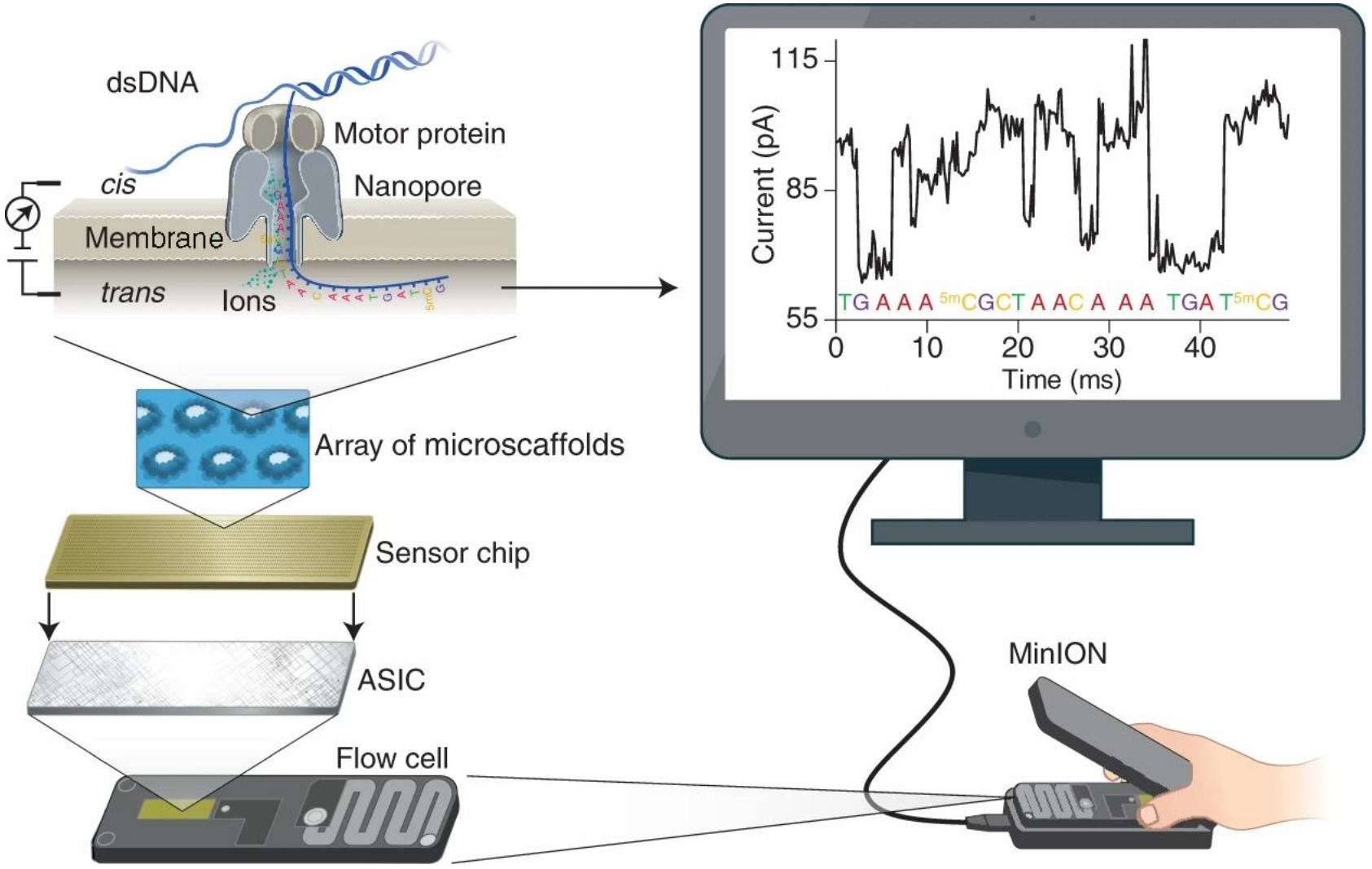
INTERPRET THE DATA *If the template strand has two or more identical nucleotides in a row, their complementary nucleotides will be added one after the other in the same flow step. How are two or more of the same nucleotide (in a row) detected in the flow-gram? (See sample on the right.) Write out the sequence of the first 25 nucleotides in the flow-gram above, starting from the left. (Ignore the very short lines.)*

Next Generation Sequencing



Nanopore sequencing

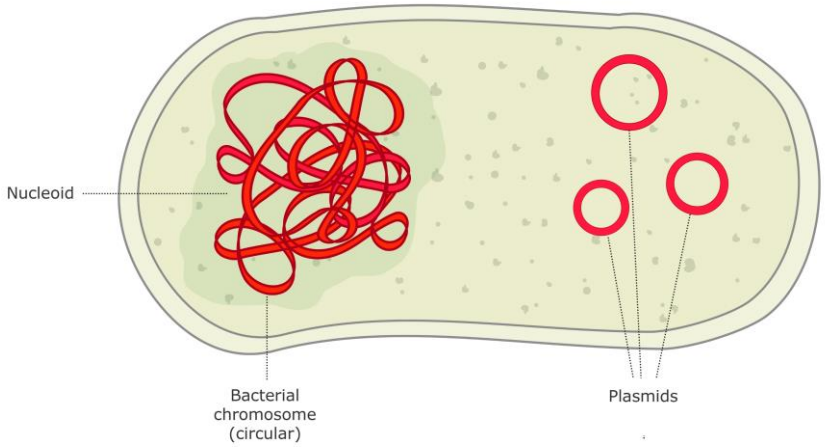
In some new methods, the DNA is neither cut into fragments nor amplified. Instead, a single, very long DNA molecule is sequenced on its own.
A single strand of a DNA molecule is moved through a very small pore (a nanopore) in a membrane, identifying the bases one by one by the distinct way each interrupts an electrical current. Here nanopore is a protein channel embedded in a lipid membrane.
(Other researchers are using artificial membranes and nanopores.) Each type of base interrupts the electrical current for a slightly different length of time. In 2015, the first nanopore sequencer went on the market; it connects to a computer via a USB port. Associated software allows immediate identification and analysis of the sequence.



DNA cloning yields multiple copies of a gene or other DNA segment

DNA cloning: isolating and amplifying a gene of interest.

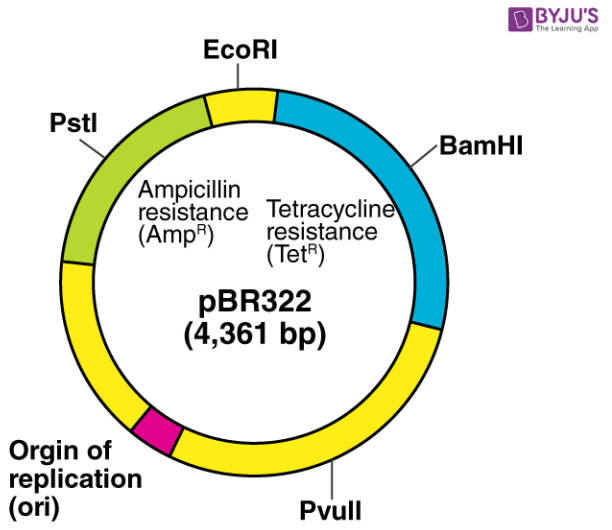
Most methods for cloning pieces of DNA in the laboratory share general features, such as the use of bacteria and their plasmids.



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Plasmids are small circular DNA molecules that replicate separately from the bacterial chromosome.

A plasmid has only a small number of genes; these genes may be useful when the bacterium is in a particular environment but may not be required for survival or reproduction under most conditions.
For example, plasmid genes may contain genes for antibiotic resistance.

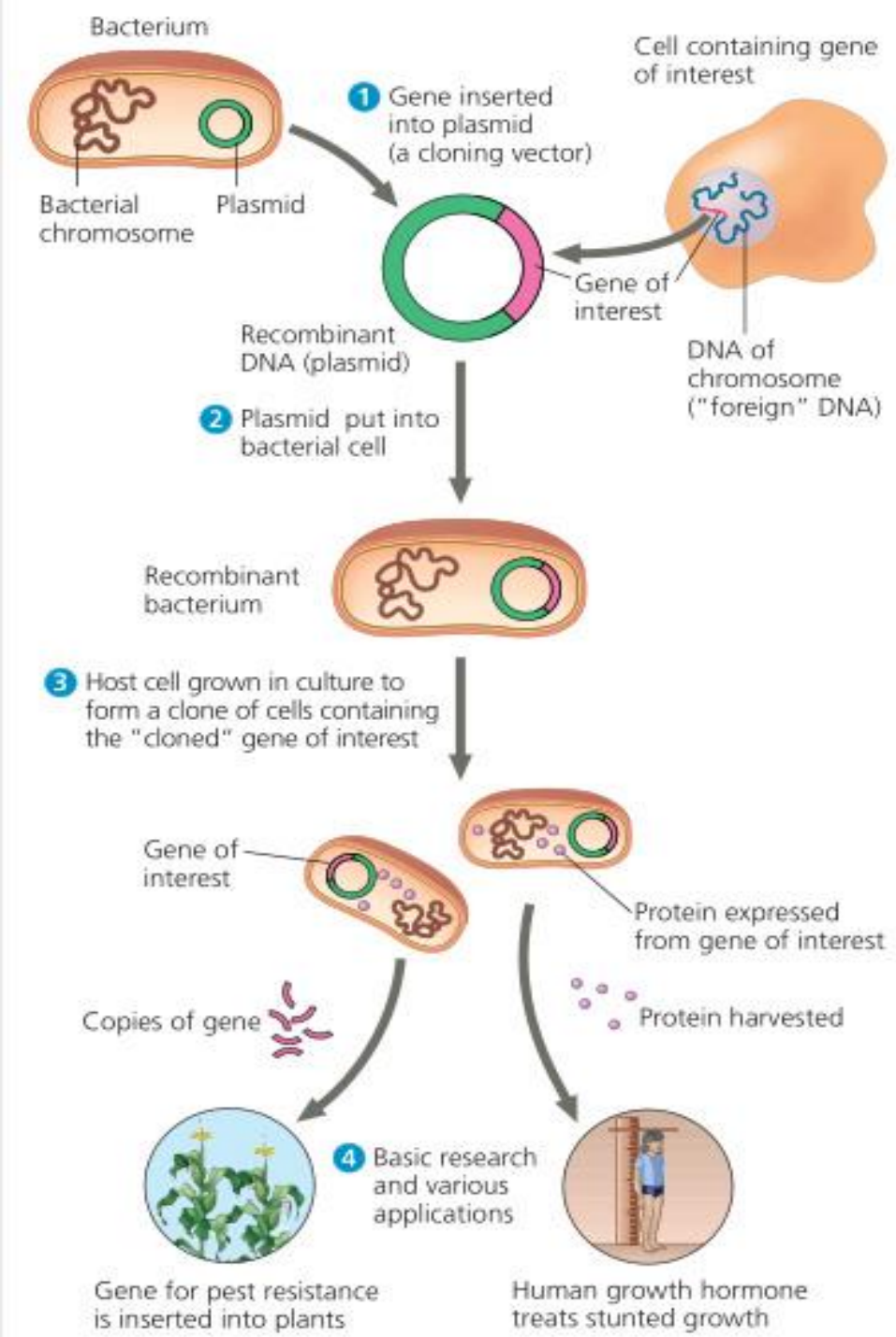


DNA Cloning and Its Applications: A Preview

To clone pieces of DNA using bacteria, scientists have isolated plasmids from bacterial cells and altered them by genetic engineering.
Researchers insert DNA they want to study (“foreign” DNA) into the plasmid.

The resulting plasmid is now a **recombinant DNA molecule**, a molecule containing DNA from two different sources, very often different species.

Gene cloning involves using bacteria to make multiple copies of a gene.
The recombinant plasmid is inserted into a bacterial cell.
Reproduction in the bacterial cell results in cloning of the plasmid including the foreign DNA.
This results in the production of multiple copies of a single gene.



Gene cloning is useful for two basic purposes:

- to make many copies of or amplify, a particular gene
- to produce a protein product from it.

- For example, a resistance gene present in one crop species might be cloned and transferred into plants of another species. (Such organisms are called genetically modified organisms, or GMOs)



Gene is used to alter bacteria for cleaning up toxic waste



Protein dissolves blood clots in heart attack therapy

Using Restriction Enzymes to Make Recombinant DNA

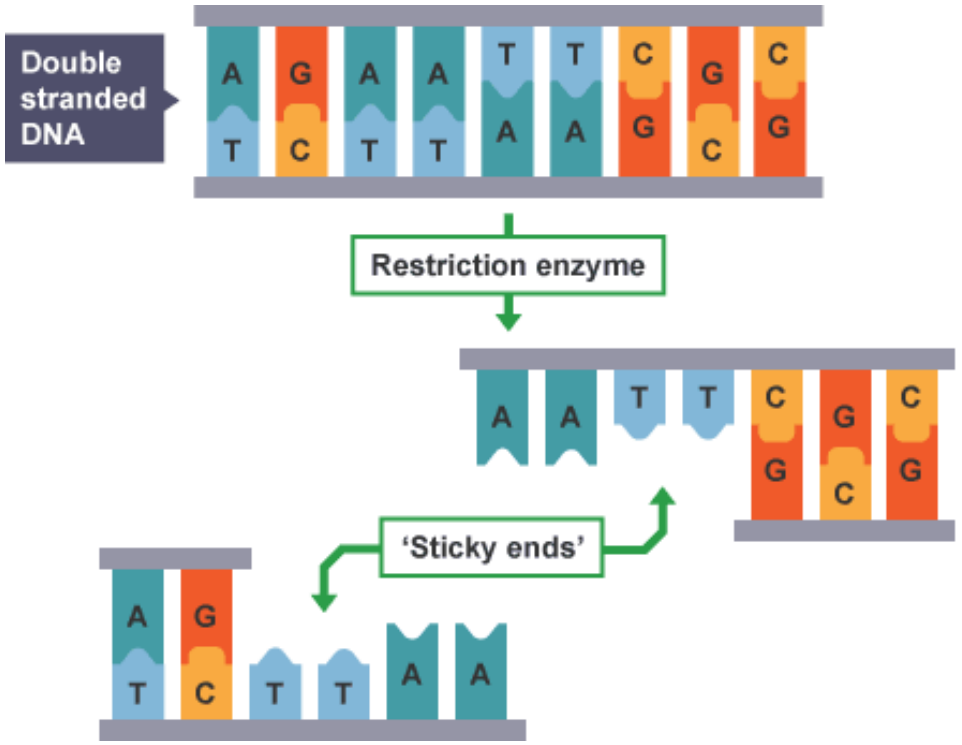
- **Bacterial restriction enzymes (restriction endonucleases)** cut DNA molecules at specific DNA sequences called restriction sites.

- A **restriction enzyme** usually makes many cuts, yielding restriction fragments.

Hundreds of different restriction enzymes have been identified and isolated.

Each restriction enzyme is very specific, recognizing a particular short DNA sequence, or restriction site, and cutting both DNA strands at precise points within this restriction site.

- The DNA of a bacterial cell is protected from the cell's own restriction enzymes by the addition of methyl groups (—CH3) to adenines or cytosines within the sequences recognized by the enzymes.
- The most useful restriction enzymes cut DNA in a staggered way, fragments with "sticky ends" that bond with complementary sticky ends of other fragments.
- Since restriction enzymes always cut at the same exact DNA sequence, copies of any given DNA molecule exposed to the same restriction enzyme always yield the same set of restriction fragments.



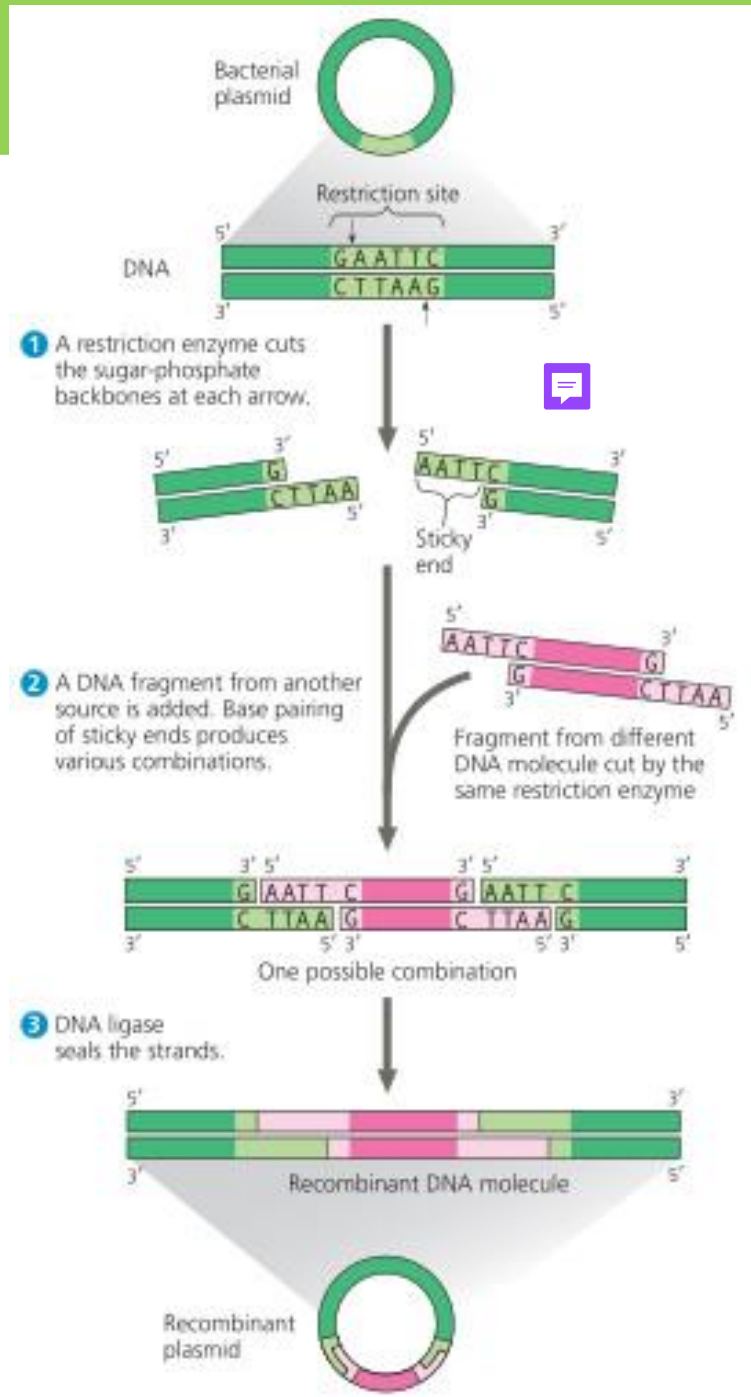
Using Restriction Enzymes to Make Recombinant DNA

Using a restriction enzyme and DNA ligase to make a recombinant DNA plasmid. The restriction enzyme in this example (called EcoRI) recognizes a single six-base-pair restriction site present in this plasmid. It makes staggered cuts in the sugarphosphate backbones, producing fragments with “sticky ends.”

Foreign DNA fragments with complementary sticky ends can base-pair with the plasmid ends; the ligated product is a recombinant plasmid. (If the two plasmid sticky ends base-pair, the original nonrecombinant plasmid is reformed.)

In gene cloning, the original plasmid is called a **cloning vector.** The foreign gene is called **the insert.**

A cloning vector is a DNA molecule that can carry foreign DNA into a host cell and replicate there.



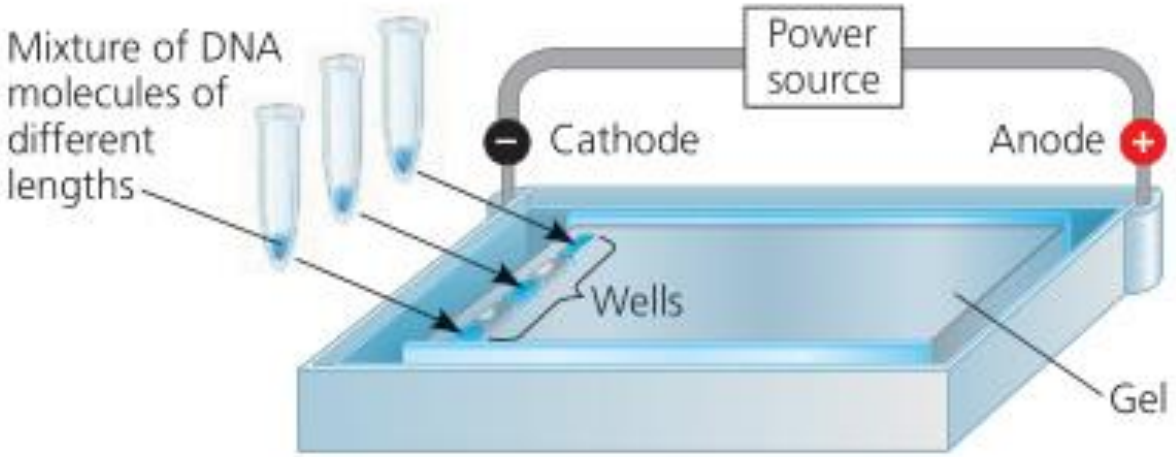
Gel Electrophoresis and Southern Blotting

One indirect method of rapidly analyzing and comparing genomes is gel electrophoresis

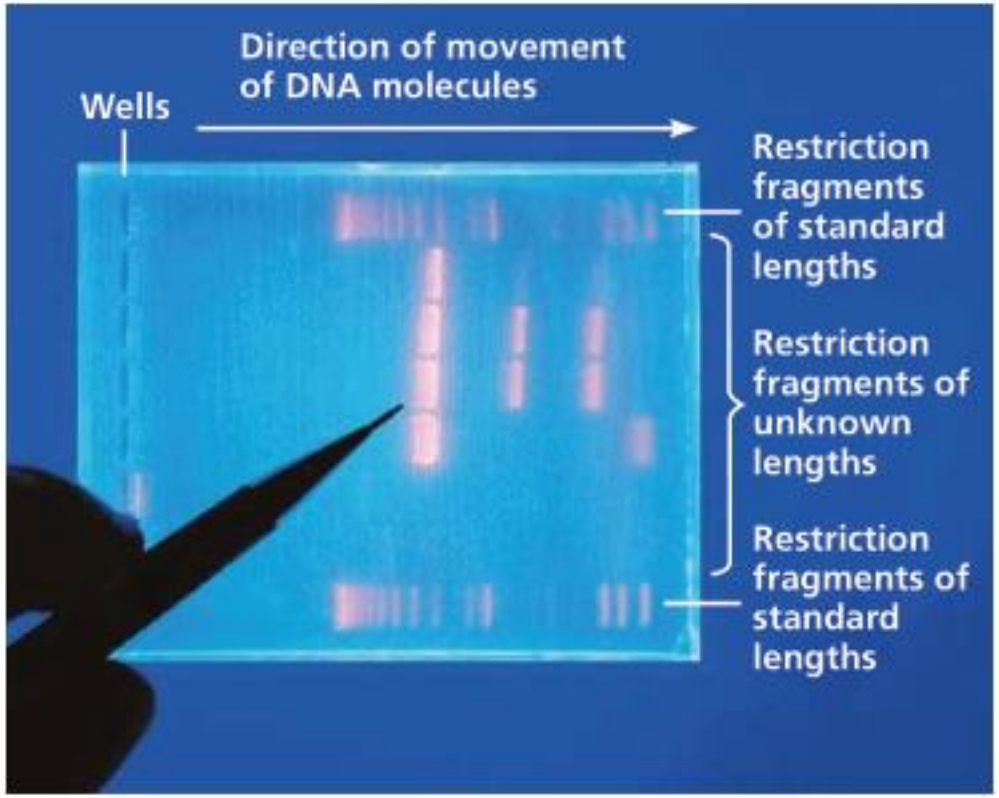
- This technique uses a gel to separate nucleic acids or proteins by size
- A current is applied that causes charged molecules to move through the gel
- Molecules are sorted into “bands” by their size
- Smaller DNA (or proteins) migrate through the gel much easier (and faster) than larger DNA/protein.
- To check the recombinant plasmid, researchers might cut the products again using the same restriction enzyme
- To separate and visualize the fragments produced, gel electrophoresis would be carried out
- This technique uses a gel made of a polymer to separate a mixture of nucleic acids or proteins based on size, charge, or other physical properties

Gel Electrophoresis

▼ Figure 20.6 Gel electrophoresis. A gel made of a polymer acts as a molecular sieve to separate nucleic acids or proteins differing in size, electrical charge, or other physical properties as they move in an electric field. In the example shown here, DNA molecules are separated by length in a gel made of a polysaccharide called agarose.



(a) Each sample, a mixture of different DNA molecules, is placed in a separate well near one end of a thin slab of agarose gel. The gel is set into a small plastic support and immersed in an aqueous, buffered solution in a tray with electrodes at each end. The current is then turned on, causing the negatively charged DNA molecules to move toward the positive electrode.

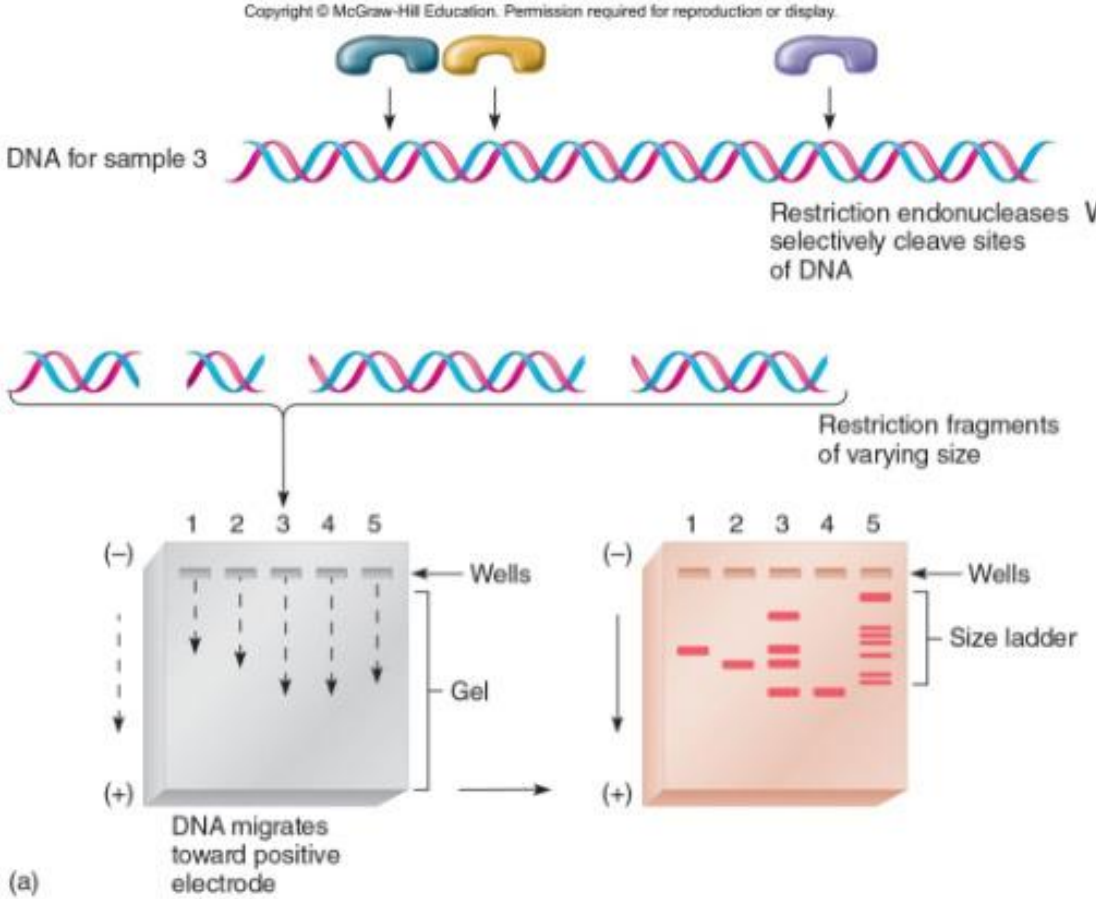


(b) Shorter molecules are slowed down less than longer molecules so shorter molecules move faster through the gel. After the current is turned off, a DNA-binding dye is added that fluoresces pink in ultraviolet (UV) light. Each pink band corresponds to many thousands of DNA molecules of the same length. The bands at the upper and lower edges of the gel are restriction fragments of standard lengths for comparison with samples of unknown length.

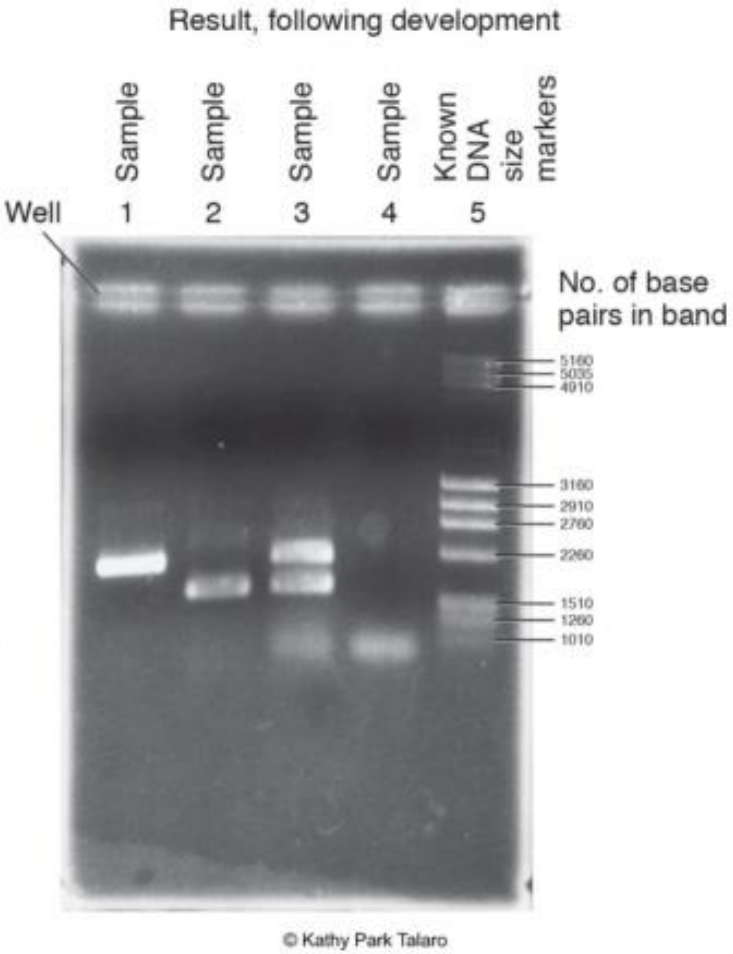
Methods for Analysis of DNA

- Negative charge of molecule causes DNA to move toward positive pole
- Rate of movement is dependent on size of fragment – larger fragments move more slowly

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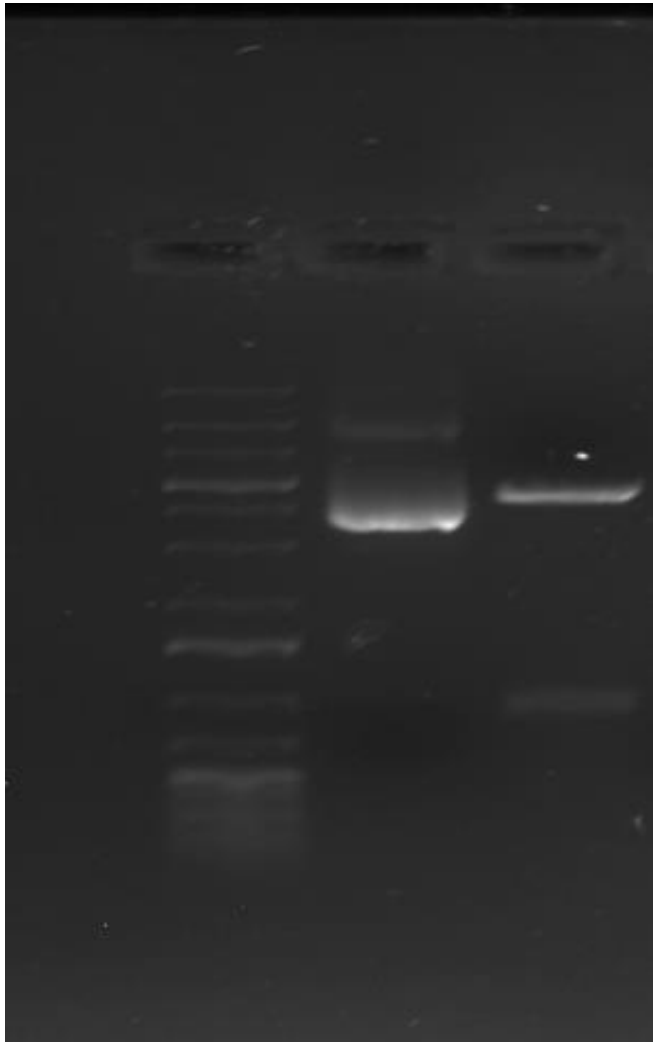


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Gel Electrophoresis and Southern Blotting

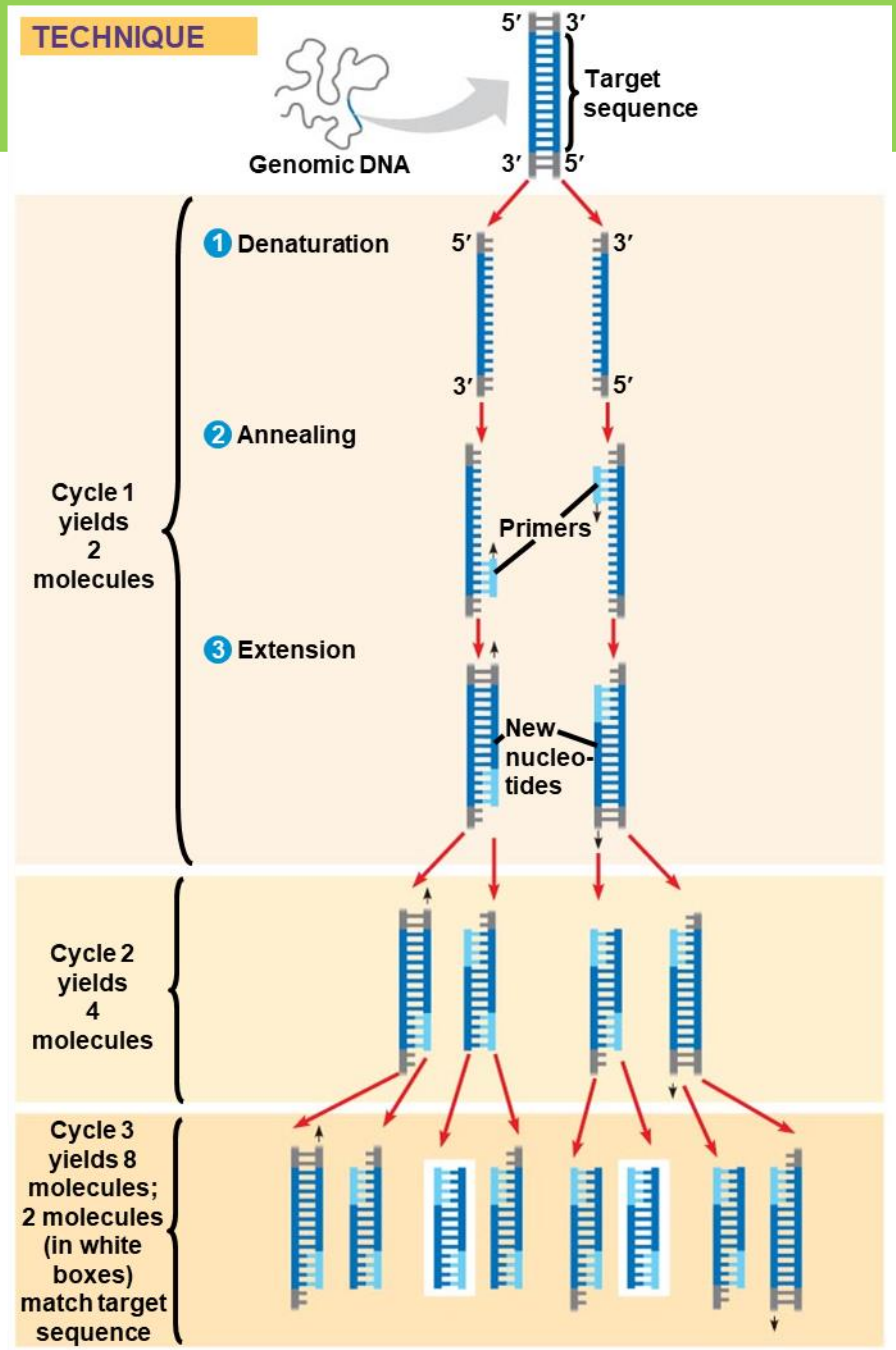
DNA ladder | Intact plasmid | RE Digested Plasmid



Amplifying DNA in Vitro: The Polymerase Chain Reaction (PCR)

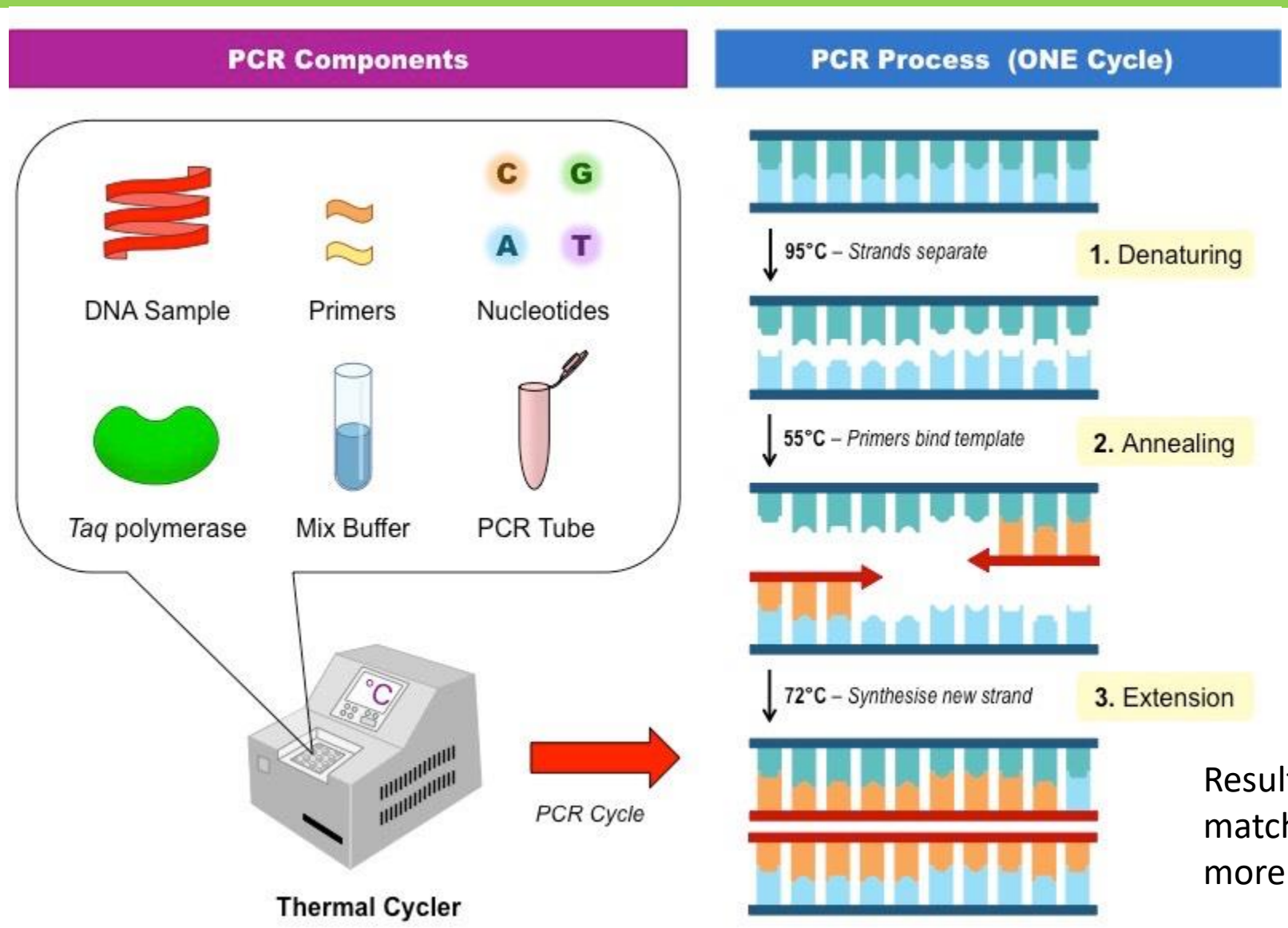
The polymerase chain reaction (PCR) can produce millions of copies of a specific target segment of DNA

- A three-step cycle—heating, cooling, and replication—brings about a chain reaction that produces an exponentially growing population of identical DNA molecules.
- The key to PCR is an unusual, heat-stable DNA polymerase called **Taq polymerase** obtained from hot-spring bacteria.



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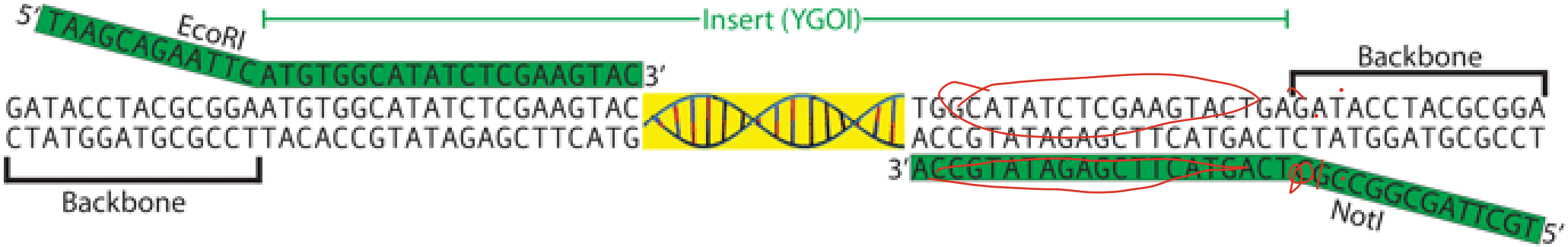
Amplifying DNA in Vitro: The Polymerase Chain Reaction (PCR)



Results After three cycles, two molecules match the target sequence exactly. After 30 more cycles, over 1 billion molecules match.

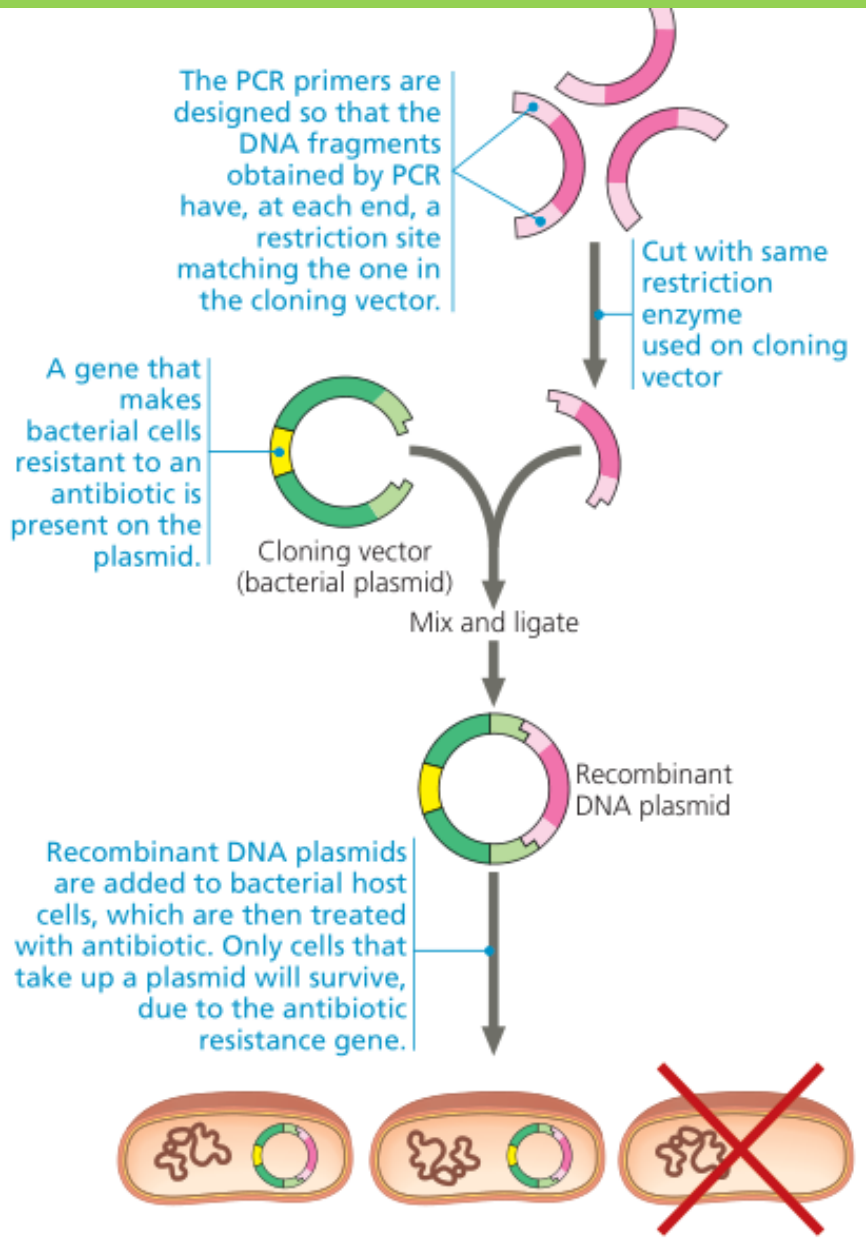
Amplifying DNA in Vitro: The Polymerase Chain Reaction (PCR)

The key to the high specificity is the pair of primers used for each PCR amplification. The primer sequences are chosen so that they hybridize only to sequences at opposite ends of the target segment, one on the 3’ end of each strand. (For high specificity, the primers must be at least 15 nucleotides long.)



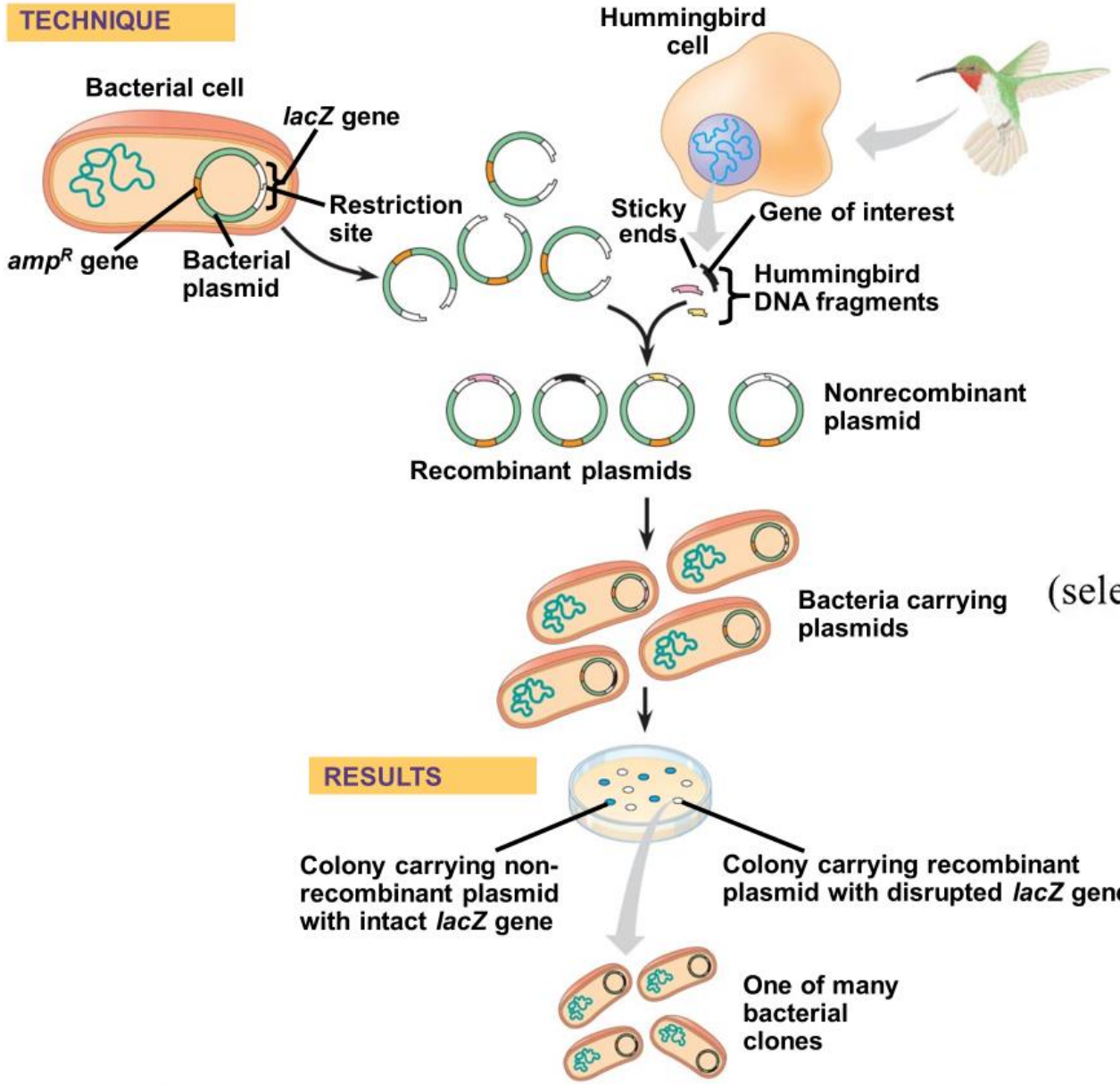
- PCR primers are synthesized to include a restriction site at each end of the DNA fragment that matches the site in the cloning vector, and the fragment and vector are cut and ligated together

Amplifying DNA in Vitro: The Polymerase Chain Reaction

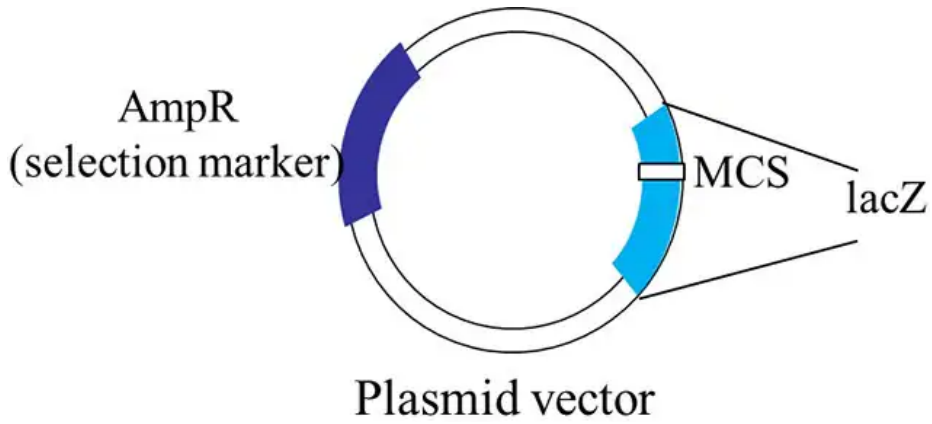


The resulting plasmids are sequenced so that those with error-free inserts canbe selected.

Fig. 20-4-4



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Expressing cloned eukaryotic genes

Cloned genes can be expressed (used to produce proteins) in either bacterial or eukaryotic cells; each option has advantages and disadvantages.

Bacterial Expression Systems

Allows getting a cloned eukaryotic gene to function in bacterial host cells.

Expression vector is a cloning vector that contains a highly active bacterial promoter just upstream of a restriction site where the eukaryotic gene can be inserted in the correct reading frame.

The bacterial host cell will recognize the promoter and proceed to express the foreign gene now linked to that promoter.

Such expression vectors allow the synthesis of many eukaryotic proteins in bacterial cells.

Another difficulty with eukaryotic gene expression in bacteria is the presence of introns in most eukaryotic genes

- Researchers can avoid this problem by using cDNA, complementary to the mRNA, which contains only exons

Eukaryotic DNA Cloning and Expression Systems

Molecular biologists can avoid eukaryote-bacterial incompatibility issues by using eukaryotic cells, such as yeasts, as hosts for cloning and expressing genes

In addition to enabling RNA splicing, eukaryotic host cells are advantageous because many eukaryotic proteins will not function unless they are modified after translation.

- Even yeasts may not possess the proteins required to modify expressed mammalian proteins properly.

- In such cases, cultured mammalian or insect cells may be used to express and study proteins

Eukaryotic DNA Cloning and Expression Systems

One method of introducing recombinant DNA into eukaryotic cells is electroporation-applying a brief electrical pulse to create temporary holes in plasma membranes

- Alternatively, scientists can inject DNA into cells using microscopically thin needles

- Once inside the cell, the DNA is incorporated into the cell's DNA by natural genetic recombination

- Eukaryotic expression is a good way to test the function of a specific protein

Analyzing Gene Expression

Analysis of when and where a gene or group of genes is expressed can provide important clues about gene function.

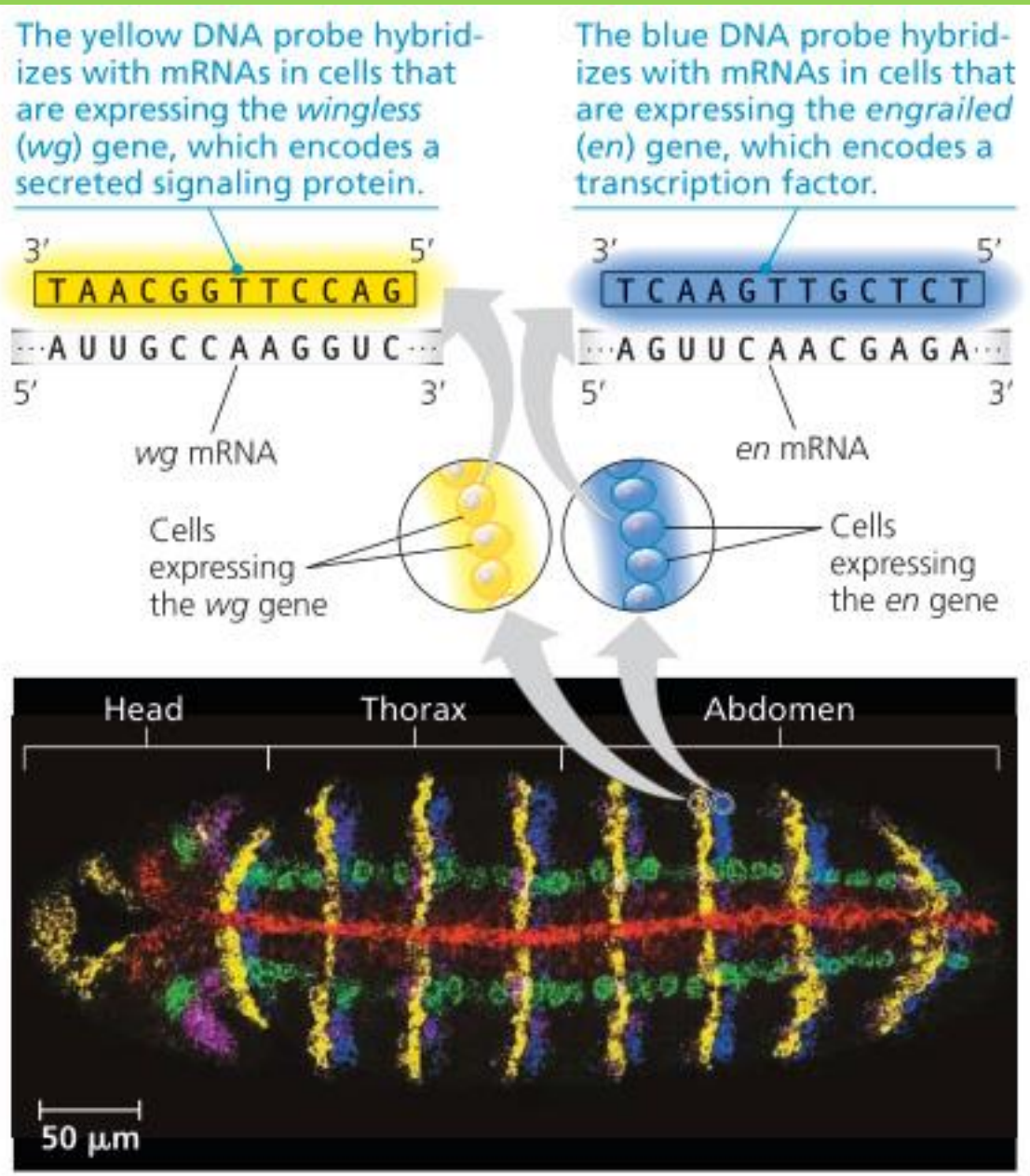
- The most straightforward way to discover which genes are expressed in certain cells is to identify the mRNAs being made.

Studying the Expression of Single Genes

- Labeled nucleic acid probes that hybridize with mRNAs can provide information about the time or place in the organism at which a gene is transcribed.
- Suppose we have cloned a gene that we suspect plays an important role in the embryonic development of Drosophila melanogaster (the fruit fly).
- To find where in the embryo is the corresponding mRNA found we use in situ (on site) hybridization of this gene to a labeled probe.

(Probes are stretches of DNA or RNA that we've attached a label to. The label allows us to see where the DNA binds either in a cell, or in a chromosome, or even in pure isolated DNA.)

in situ hybridization



This Drosophila embryo was incubated in a solution containing DNA probes for five different mRNAs, each probe labeled with a different fluorescently colored tag.
The embryo was then viewed using fluorescence microscopy.
Each color marks where a specific gene is expressed as mRNA.
The arrows from the groups of yellow and blue cells above the micrograph show a magnified view of nucleic acid hybridization of the appropriately colored probe to the mRNA. The thorax (trunk) and abdomen are made up of repeating segments.
Yellow cells (expressing the wg gene) interact with blue cells (expressing the en gene); their interaction helps establish the pattern in a body segment.

Reverse transcriptase polymerase chain reaction, or RT-PCR

- Other mRNA detection techniques may be preferable for comparing the amounts of a specific mRNA in several samples at the same time—for example, in different cell types or in embryos of different stages.

- One method that is widely used is called the reverse transcriptase polymerase chain reaction, or RT-PCR.

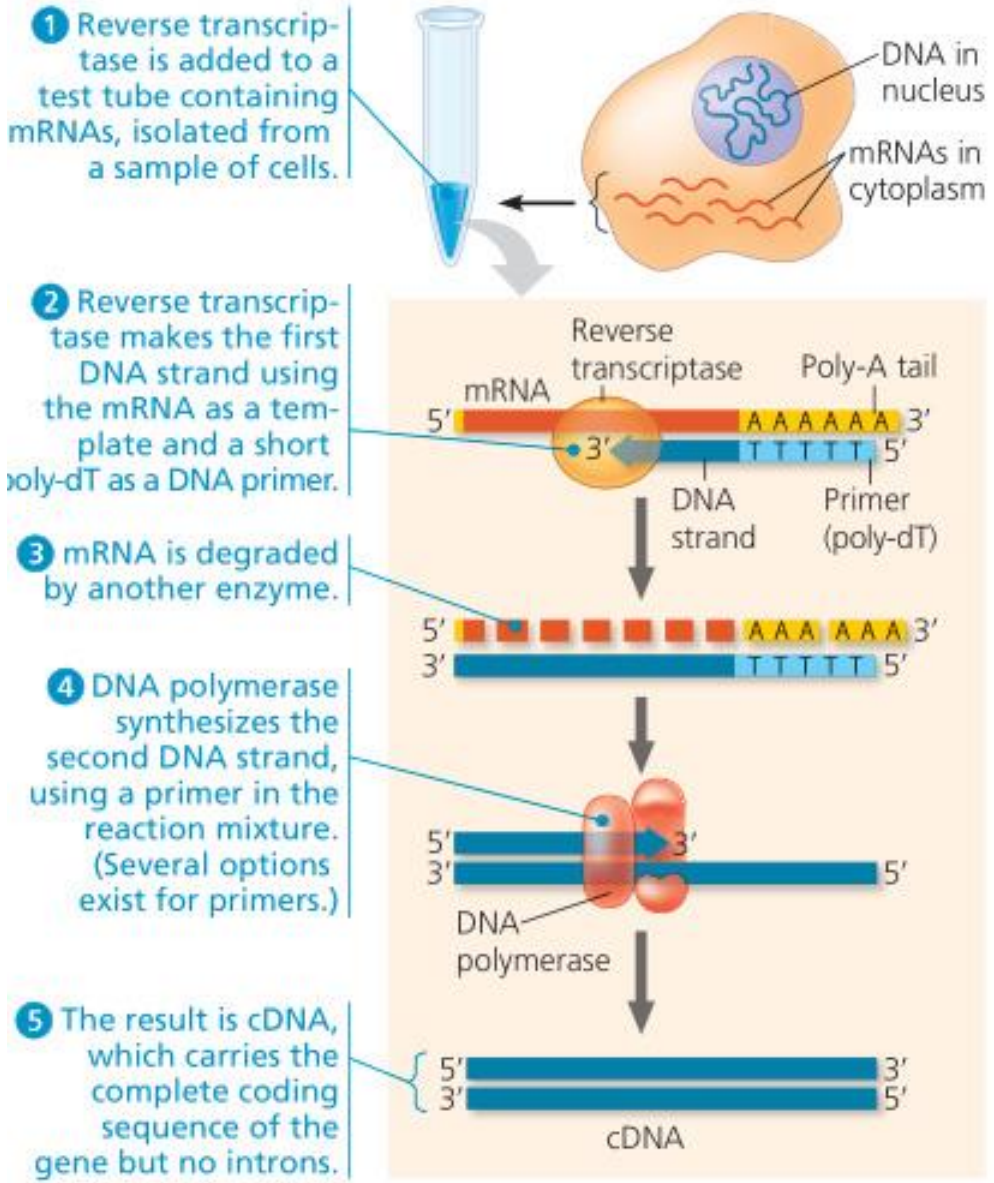
Reverse transcriptase polymerase chain reaction, or RT-PCR

First, the enzyme reverse transcriptase (from a retrovirus) is used to synthesize a complementary DNA copy (a reverse transcript) of each mRNA in the sample.

The mRNA is then degraded by addition of a specific enzyme, and a second DNA strand, complementary to the first, is synthesized by DNA polymerase. The resulting double-stranded DNA is called complementary DNA, (cDNA). (Made from mRNA, cDNA lacks introns and can be used for protein expression in bacteria, as mentioned earlier.)

For example, to analyze the timing of expression of the Drosophila gene of interest, we would first isolate all the mRNAs from different stages of Drosophila embryos and make cDNA from each stage.

▼ Figure 20.10 Making complementary DNA (cDNA) from eukaryotic genes. Complementary DNA is made in a test tube using mRNA as a template for the first strand. Only one mRNA is shown after step 1, but the final collection of cDNAs would reflect all the mRNAs present in the cell.

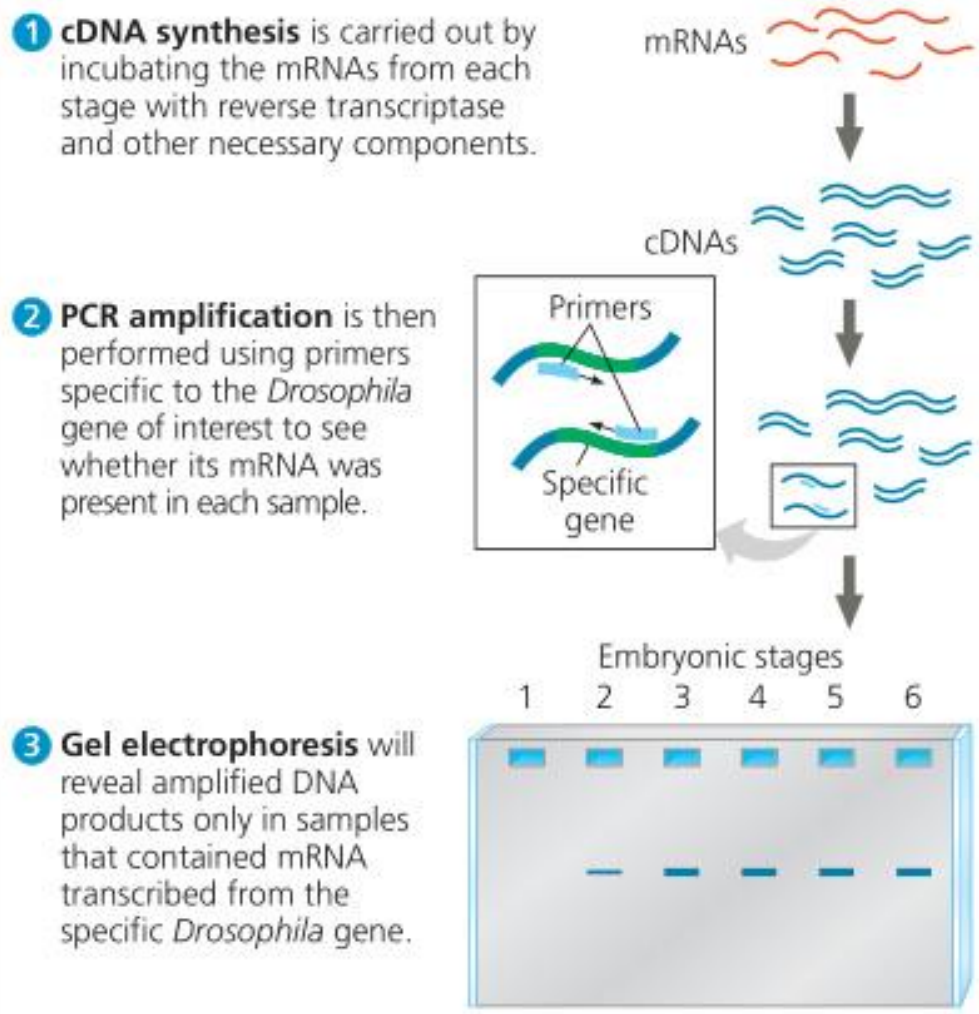


RT-PCR Analysis of the Expression of Single Genes

Application RT-PCR uses the enzyme reverse transcriptase (RT) in combination with PCR and gel electrophoresis. RT-PCR can be used to compare gene expression between samples—
for instance, in different embryonic stages, in different tissues, or in the same type of cell under different conditions.

However, accurate measurement of mRNA levels requires newer, more quantitative PCR machines that use a dye that fluoresces only when bound to a double-stranded PCR product. This technique, called quantitative RT-PCR (qRT-PCR), can detect the light and measure the PCR product, thus avoiding the need for electrophoresis while providing quantitative data, a distinct advantage.

Technique In this example, samples containing mRNAs from six embryonic stages of *Drosophila* were analyzed for a specific mRNA as shown below. (In steps 1 and 2, the mRNA from only one stage is shown.)



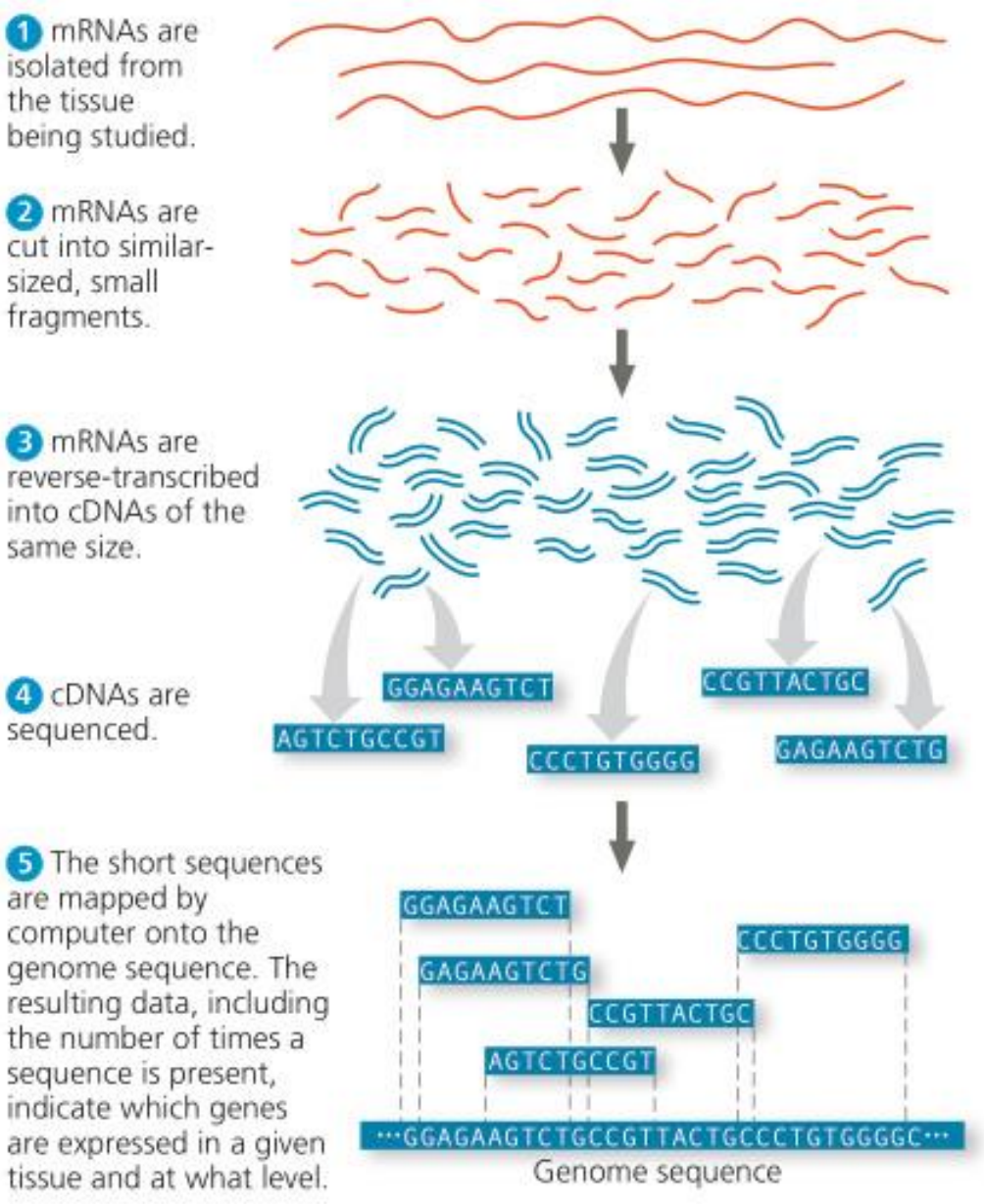
Results The mRNA for this gene is expressed from stage 2 through stage 6. The size of the amplified fragment (shown by its position on the gel) depends on the distance between the primers that were used (not on the size of the mRNA).

Studying the Expression of Interacting Groups of Genes

It is possible to study the expression of large groups of genes—an approach called the systems approach.
Researchers use what is known about the whole genome to investigate which genes are transcribed in different tissues or at different stages of development. One aim is to identify networks of gene expression across an entire genome.

To accomplish such genome-wide expression studies, the expressed mRNA's from different tissues or different embryonic stages are isolated, fragmented and reverse transcribed into cDNA which is then sequenced. A computer program reassembles them, either mapping them onto the genome of the species in question (when available) or simply putting the fragments in order from scratch based on overlapping sequences of multiple RNAs. This straightforward method is called RNA sequencing, or RNA-seq.

▼ Figure 20.12 Use of RNA sequencing (RNA-seq) to analyze expression of many genes. RNA-seq yields a wide range of information about expression of genes, including their level of expression.



Several characteristics make RNA-seq a powerful technique.

- it is not based on hybridization with a labeled probe, so it doesn't depend on knowing genomic sequences.
- it can measure levels of expression over a very wide range.
- a careful analysis provides a wealth of information about expression of a particular gene, such as relative levels of alternatively spliced mRNAs.

In most cases, however, expression of individual genes still needs to be confirmed by RT-PCR.

DNA microarray assays

DNA microarray assay is an older method of genome-wide expression studies

It compares patterns of gene expression in different tissues, at different times, or under different conditions

A DNA microarray consists of tiny amounts of a large number of single-stranded DNA fragments representing different genes fixed to a glass slide in a tightly spaced array, or grid, of dots.

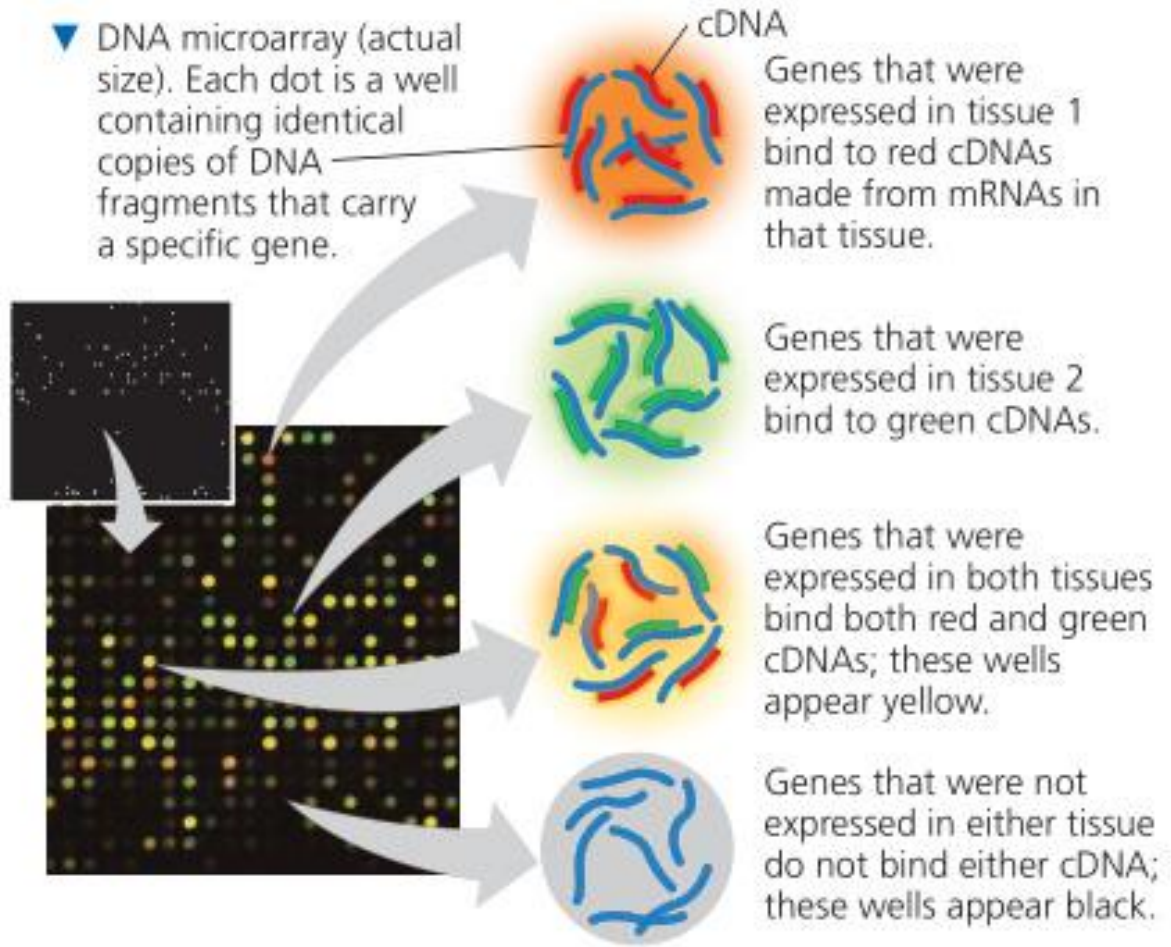
The mRNAs from the cells being studied are reverse-transcribed into cDNAs and a fluorescent label is added so the cDNAs can be used as probes on the microarray.

The resulting pattern of colored dots, shown in an actual-size microarray in Figure, reveals the dots to which each probe was bound and thus the genes that are expressed in the cell samples being tested.

DNA microarray assays

By uncovering gene interactions and providing clues
to gene function, DNA microarray assays and RNA-seq may
contribute to a better understanding of diseases and suggest
new diagnostic techniques or therapies. For instance, comparing
patterns of gene expression in breast cancer tumors
and noncancerous breast tissue has already resulted in more
informed and effective treatment protocols

▼ Figure 20.13 Use of microarrays to analyze expression of many genes. Researchers extracted mRNAs from two different human tissues and synthesized two sets of cDNAs, fluorescently labeled red (tissue 1) or green (tissue 2). Labeled cDNAs were hybridized with a microarray containing 5,760 human genes (about 25% of human genes), part of which is shown in the enlargement. Red indicates that the gene in that well was expressed in tissue 1, green in tissue 2, yellow in both, and black in neither. The fluorescence intensity at each spot indicates the relative expression of the gene.



Determining Gene Function

One way to determine function is to disable the gene and observe the consequences.

Using in vitro mutagenesis, mutations are introduced into a cloned gene, altering or destroying its function.

When the mutated gene is returned to the cell, the normal gene's function might be determined by examining the mutant's phenotype.

- The mutated gene is returned to a cell in such a way that it disables ("knocks out") the normal cellular copies of the same gene.

- If the introduced mutations alter or destroy the function of the gene product, the phenotype of the mutant cell may help reveal the function of the missing normal protein.

Using CRISPR to Edit Genes and Correct Disease-Causing Mutations

- **Gene editing:** altering genes in a specific, predictable way.
- CRISPR-Cas9 system is a powerful tool for gene editing in living cells.
- CRISPR is a bacterial immune defense against viral infections.
- Cas9 acts together with “guide RNA” made from the CRISPR region of the bacterial genome.

CRISPR utilizes a guiding RNA molecule to target a specific sequence in the genome, directing the Cas9 enzyme to cut the DNA at that location.
This cut triggers the cell's natural repair machinery, leading to either gene knockout, repair with a modified sequence, or the insertion of new genetic material.
CRISPR-Cas9 has wide-ranging applications in genetic research, agriculture, and medicine, offering the potential to treat genetic disorders and create genetically modified organisms with desired traits.

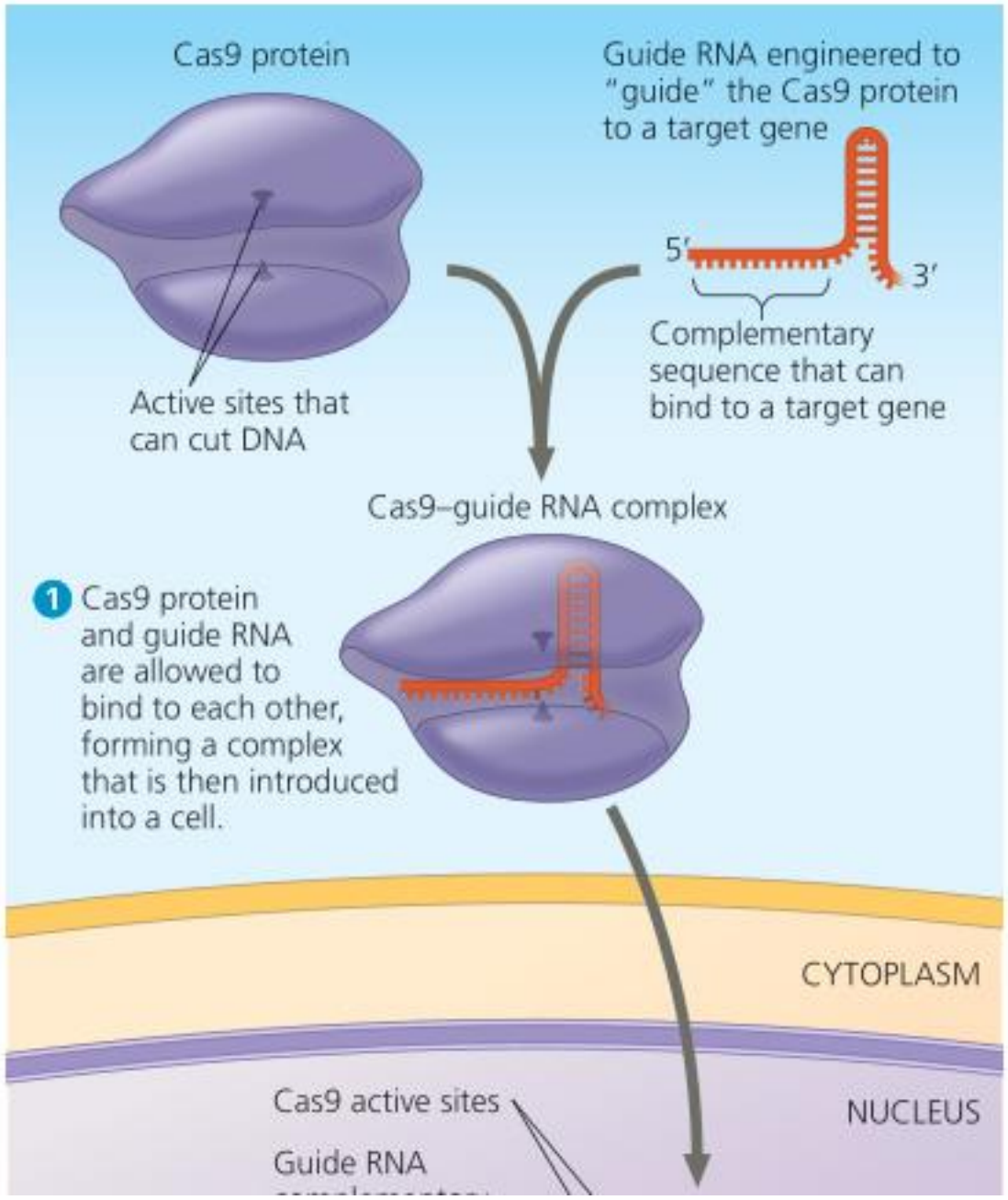
Editing Genes and Genomes

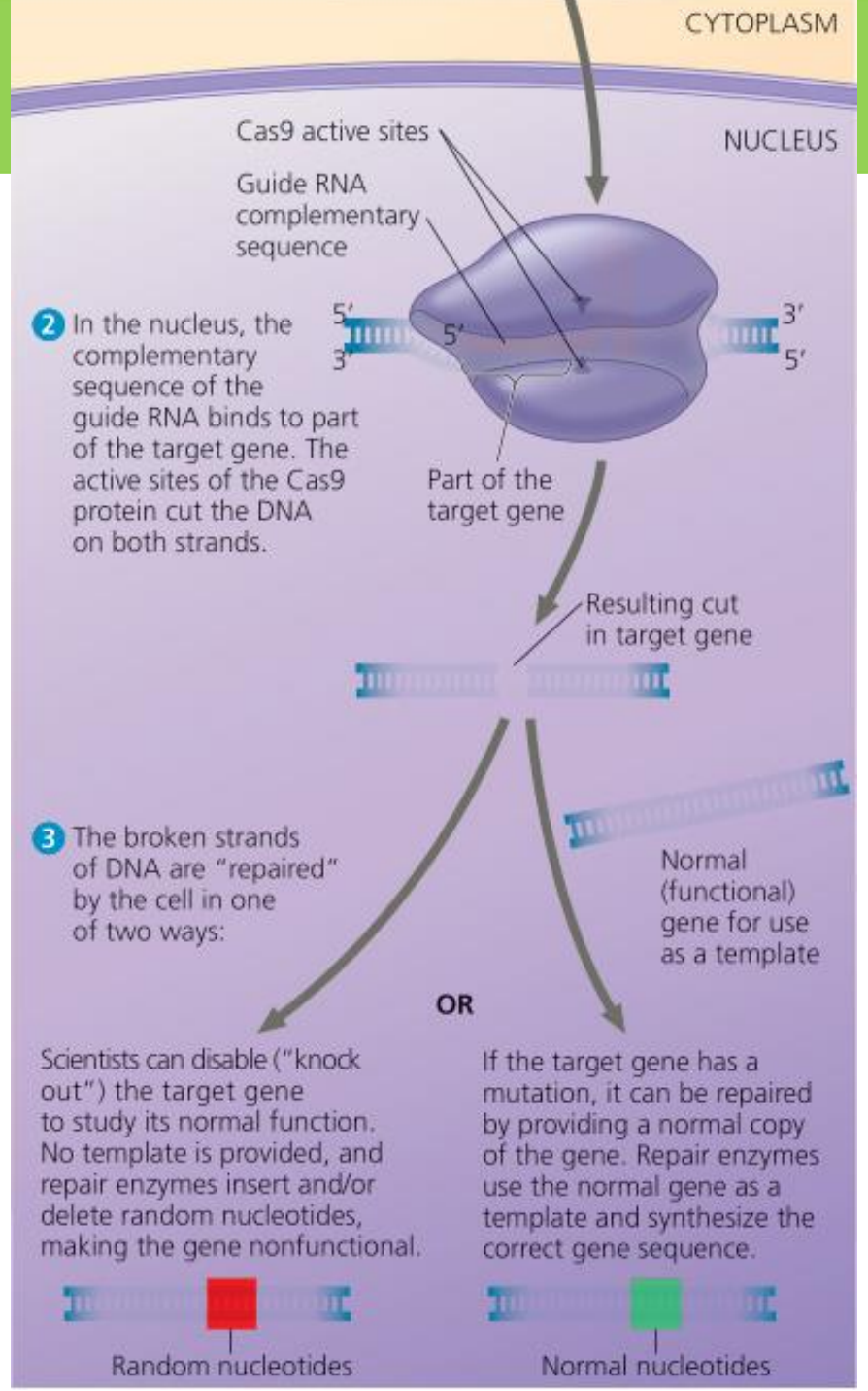
- CRISPR-Cas9 system, is a highly effective way for researchers to knock out (disable) a given gene in order to study what that gene does.

- Modifications of the technique allow researchers to repair a gene that has a mutation. This approach is used for gene therapy.

- In another application of the CRISPR-Cas9 system, scientists are attempting to address the global problem of insect-borne diseases by altering genes in insects so that, for example, they cannot transmit disease. Also the new allele can be engineered so that it is much more highly favored for inheritance than the wild-type allele. This technology is called a gene drive because the biased inheritance of the engineered gene during reproduction rapidly “drives” the new allele through the population.

▼ Figure 17.28 Gene editing using the CRISPR-Cas9 system.





Other Methods for Studying Gene Function

- Gene expression can also be silenced using **RNA interference (RNAi)**.

- Synthetic double-stranded RNA molecules matching the sequence of a particular gene are used to break down or block the gene's mRNA.

- This method is quicker than using the CRISPR-Cas9 system, but it leads to only a temporary reduction of gene expression rather than a permanent
- gene knockout or alteration.

Other Methods for Studying Gene Function

- In humans, ethical considerations prohibit knocking out genes to determine their functions. An alternative approach is to analyze the genomes of large numbers of people with a certain phenotypic condition or disease, such as heart disease or diabetes, to try to find differences they all share compared with people without that condition. The logic is that these differences may be associated with one or more malfunctioning genes and thus in a sense are naturally occurring gene knockouts.

- Genetic disorders can be tested for using genetic markers that are linked to the disease-causing allele
- Just like the coding sequences of genes, noncoding DNA at a specific locus on a chromosome may exhibit small nucleotide differences among individuals. Variations in coding or noncoding DNA sequences among a population are called polymorphisms (from the Greek for “many forms”).

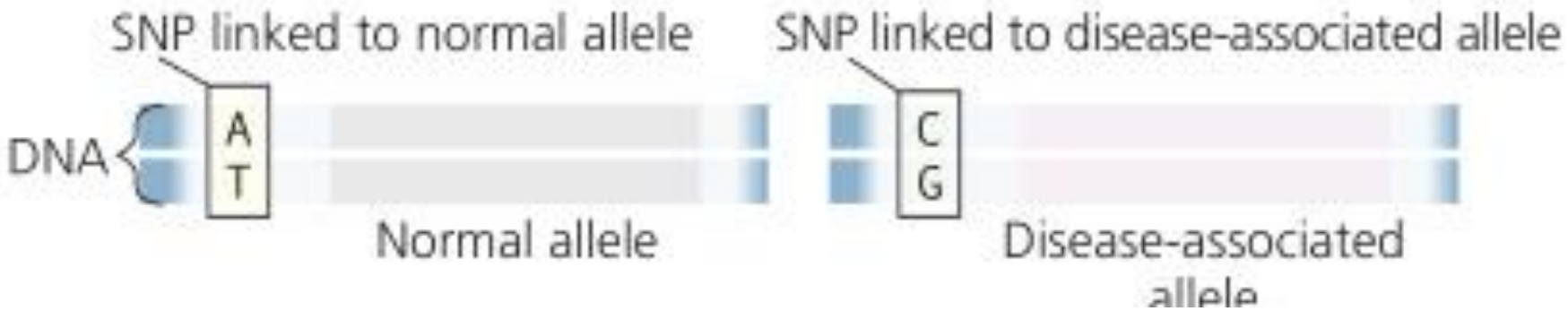
- **Single nucleotide polymorphisms** (SNPs) are useful genetic markers

- These are single base-pair sites that vary in a population

- When a restriction enzyme is added, SNPs result in DNA fragments with different lengths, or restriction fragment length polymorphism (RFLP)

Other Methods for Studying Gene Function

▼ Figure 20.15 Single nucleotide polymorphisms (SNPs) as genetic markers for disease-associated alleles. This diagram depicts the same region of the genome from two groups of individuals, one group having a particular disease or condition with a genetic basis. Unaffected people have an AT pair at a given SNP locus, while affected people have a CG pair there. Once the allele is confirmed as being associated with the disease in question, the SNP that varies in this way can be used as a marker for the disease-associated allele.



Cloned organisms and stem cells are useful for basic research and other applications

- Organismal cloning produces one or more organisms genetically identical to the “parent” that donated the single cell.
- The current interest in organismal cloning is primarily due to its ability to generate stem cells.
- A stem cell is a relatively unspecialized cell that can both reproduce itself indefinitely and, under appropriate conditions, differentiate into specialized cells of one or more types. Stem cells have great potential for regenerating damaged tissues.

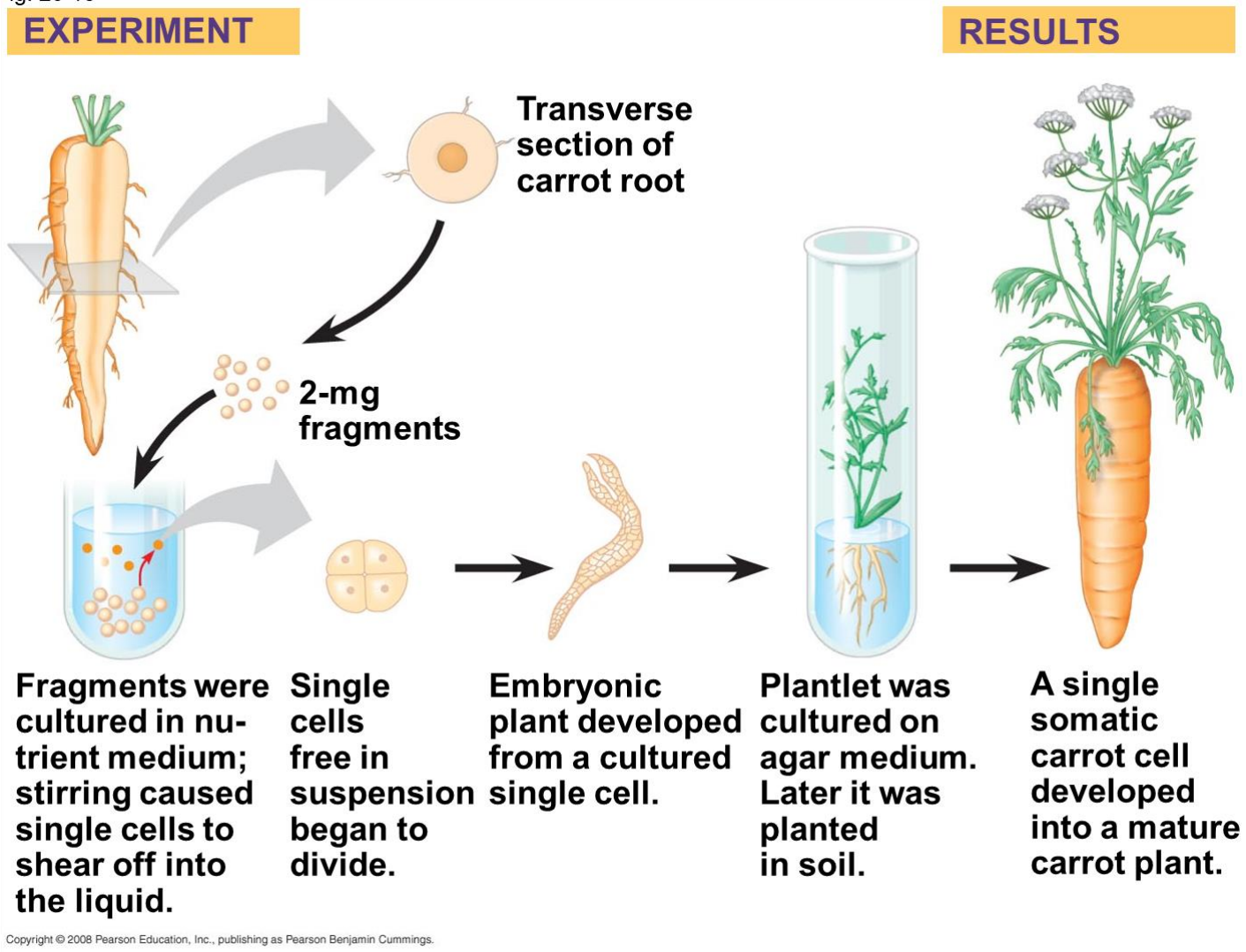
▶ Figure 20.17
Reproductive cloning of a mammal by nuclear transfer. Dolly, shown here as a lamb, has a very different appearance from her surrogate mother, standing beside her.



Cloning Plants: Single-Cell Cultures

- In plants, cells can dedifferentiate and then give rise to all the specialized cell types of the organism
- A totipotent cell, such as this, is one that can generate a complete new organism
- Plant cloning is used extensively in agriculture

Fig. 20-16



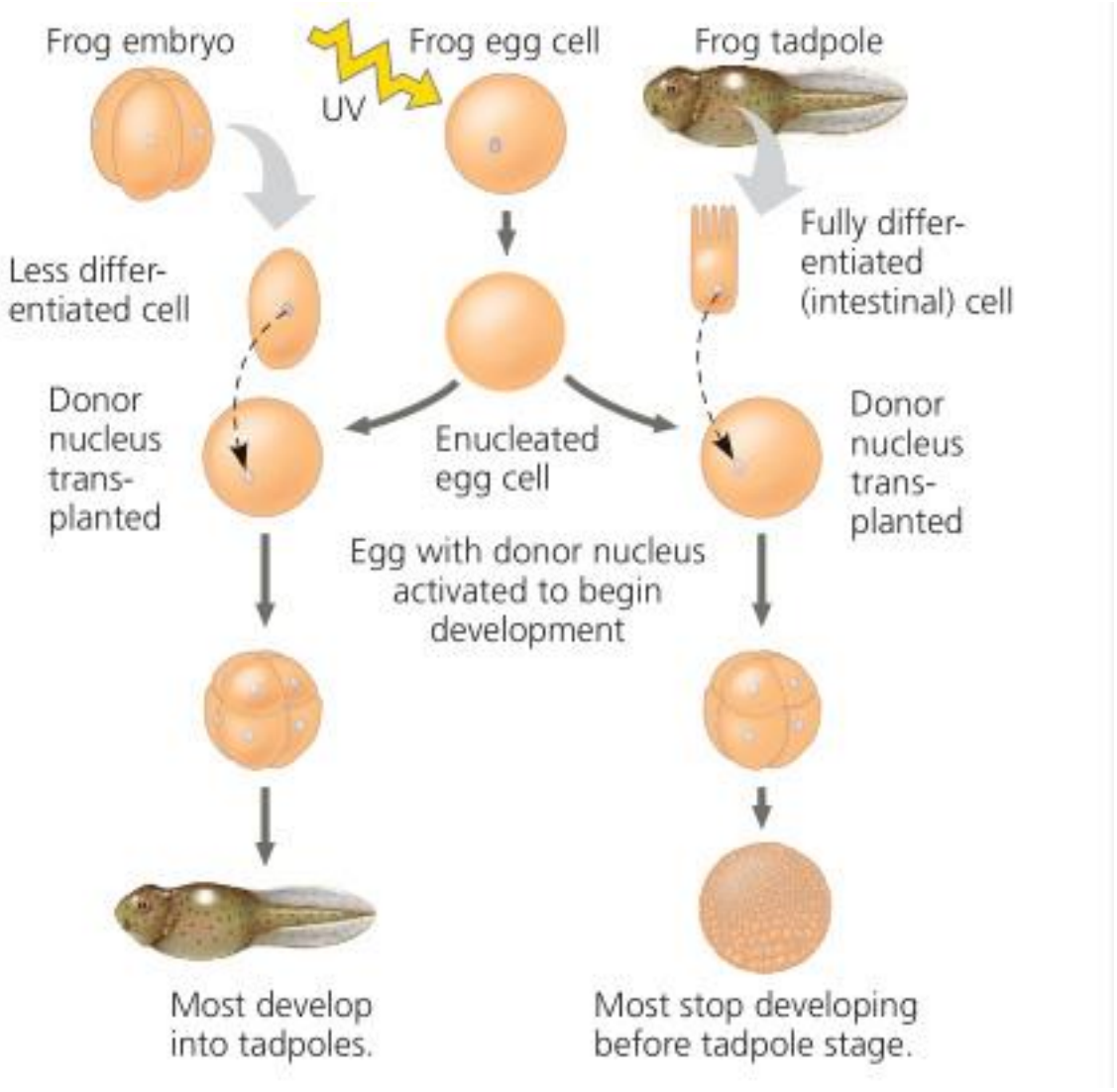
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Cloning Animals: Nuclear Transplantation (somatic cell nuclear transfer)

- In nuclear transplantation, the nucleus of an unfertilized egg cell or zygote is replaced with the nucleus of a differentiated cell

- Experiments with frog embryos have shown that a transplanted nucleus can often support normal development of the egg

- However, the older the donor nucleus, the lower the percentage of normally developing tadpolespresumably because of changes in the nucleus.



Reproductive Cloning of Mammals

- In 1997, researchers in Scotland announced the birth of Dolly, a lamb cloned from an adult sheep by nuclear transfer from a differentiated mammary gland cell

- We have learned from such experiments that cloned animals of the same species are not always identical.

- In the case of CC, fur pattern is different from the donor because of random X chromosome inactivation.

- Clearly, environmental influences and random phenomena play a significant role during development.

▼ **Figure 20.18 CC ("Carbon Copy"), the first cloned cat (right), and her single parent.** Rainbow (left) donated the nucleus in a cloning procedure that resulted in CC. However, the two cats are not identical: Rainbow is a classic calico cat with orange patches on her fur and has a "reserved personality," while CC has a gray and white coat and is more playful.

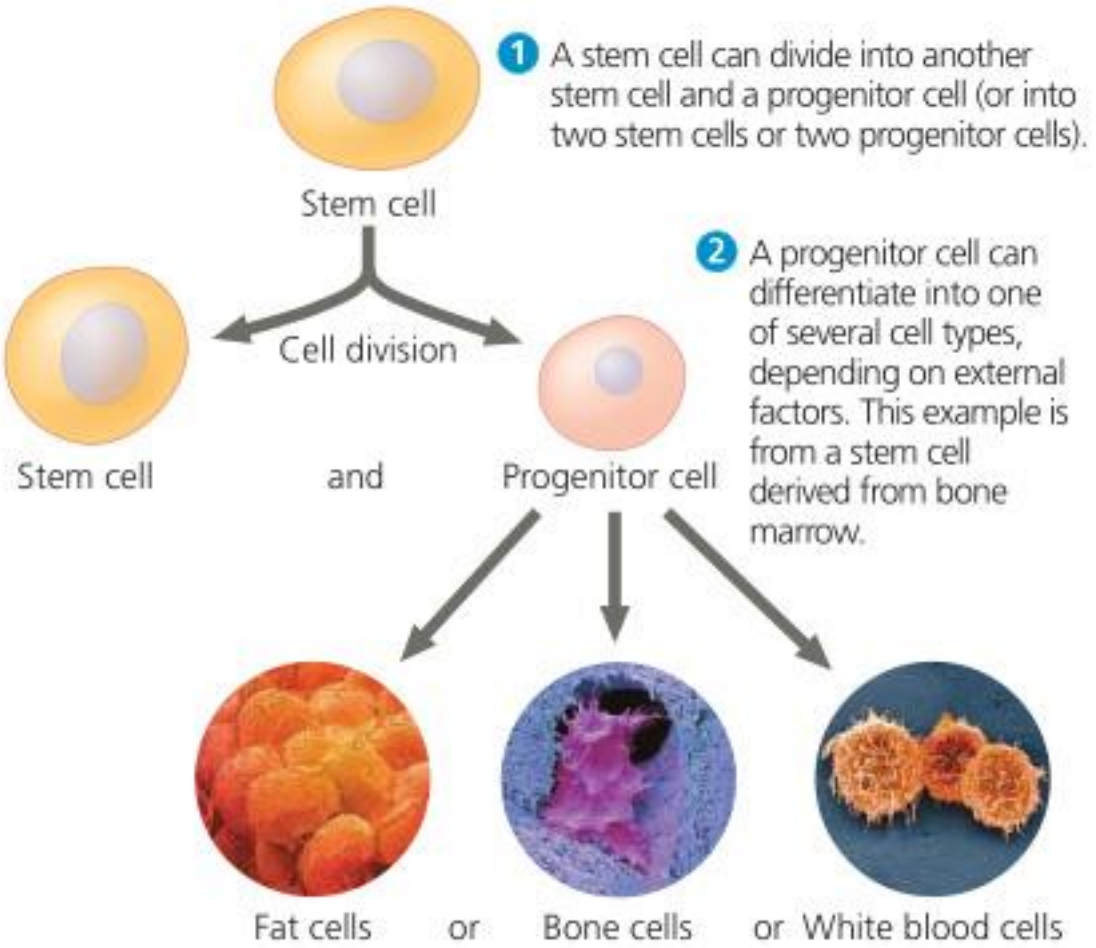


Embryonic and Adult Stem Cells

- **A stem cell** is a relatively unspecialized cell that can reproduce itself indefinitely, or under certain conditions can differentiate into one or more types of specialized cells.
- Many early animal embryos contain stem cells capable of giving rise to differentiated cells of any type.
- Stem cells have great potential for regenerating damaged tissues.

- The main reason researchers have been trying to clone human embryos is not for reproduction, but for the production of stem cells to treat human diseases.
- When the main aim of cloning is to produce embryonic stem (ES) cells to treat disease, the process is called **therapeutic cloning**.

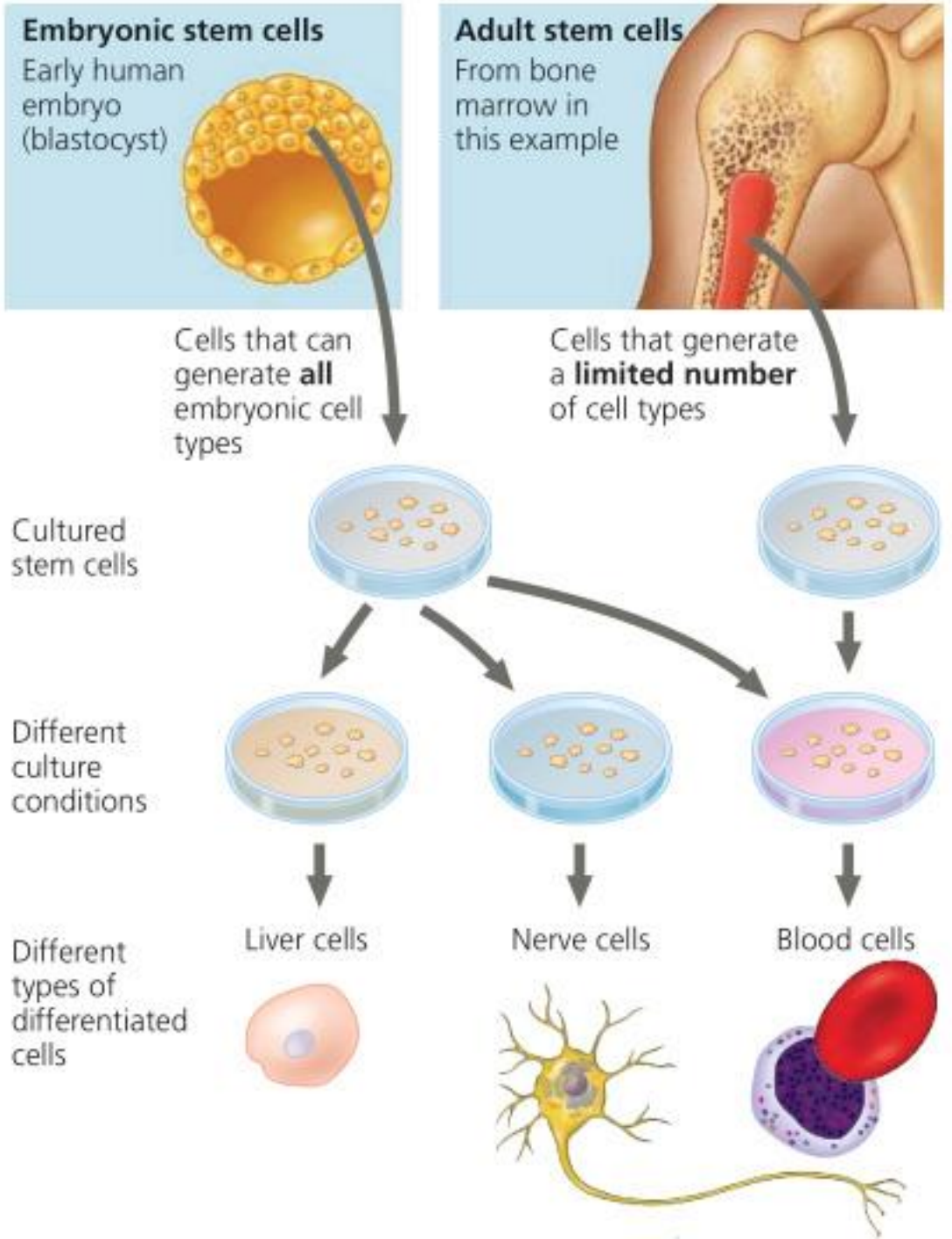
▼ Figure 20.19 How stem cells maintain their own population and generate differentiated cells.



Embryonic and Adult Stem Cells

Animal stem cells, which can be isolated from early embryos or adult tissues and grown in culture, are self-perpetuating, relatively undifferentiated cells.
Embryonic stem cells are easier to grow than adult stem cells and can theoretically give rise to all types of cells in an organism.
The range of cell types that can arise from adult stem cells is not yet fully understood.

ES cells are **pluripotent**, capable of differentiating into many different cell types.



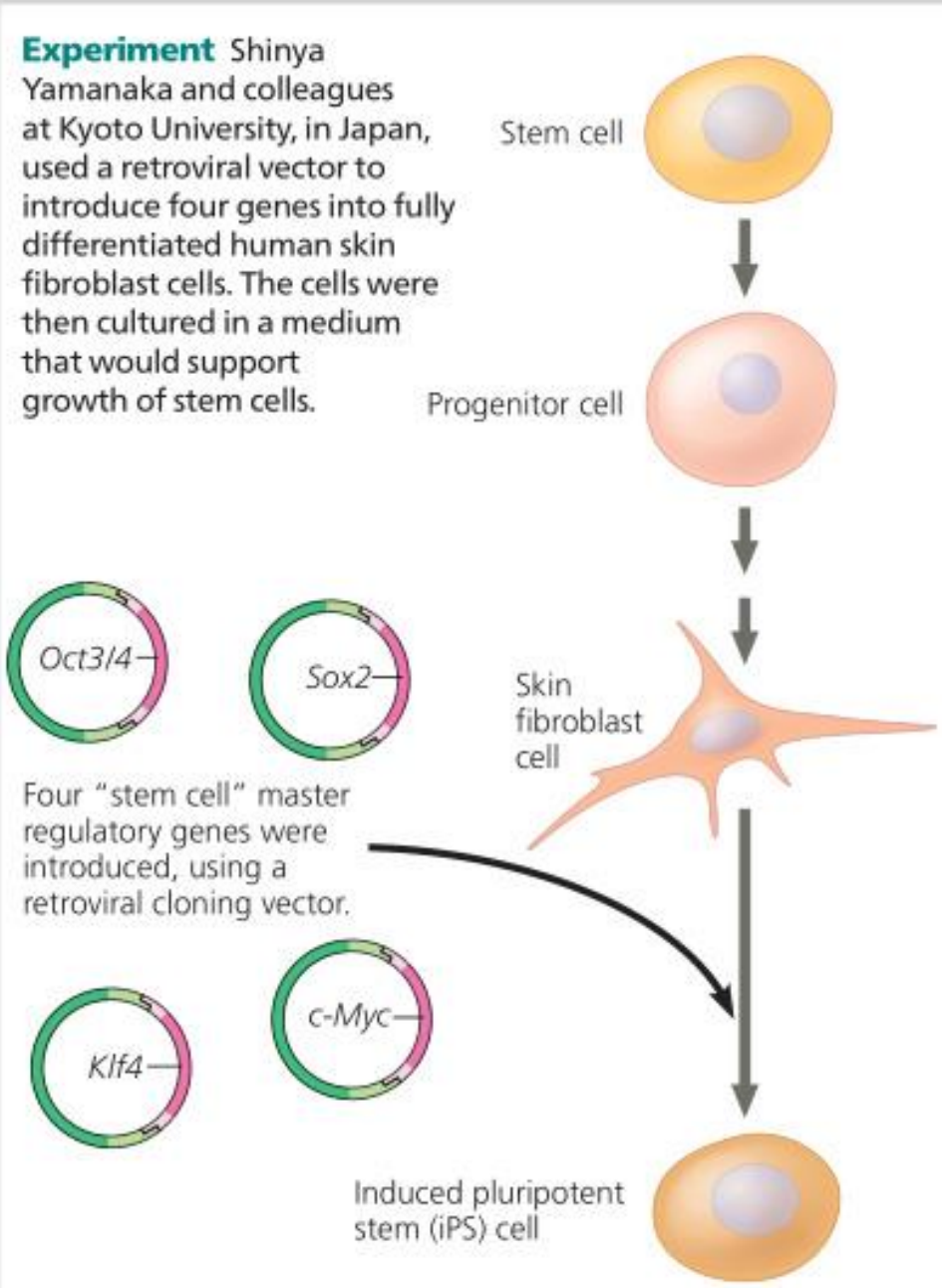
Induced Pluripotent Stem (iPS) Cells

Reprogramming fully differentiated cells to act like ES cells is possible.

Differentiated cells can be transformed into a type of ES cell by using a modified retrovirus to introduce extra, cloned copies of four “stem cell” master regulatory genes.

The “deprogrammed” cells are known as induced pluripotent stem (iPS) cells because, in using this fairly simple laboratory technique to return them to their undifferentiated state, pluripotency has been restored.

Can a fully differentiated human cell be "deprogrammed" to become a stem cell?



Results Two weeks later, the cells resembled embryonic stem cells in appearance and were actively dividing. Their gene expression patterns, gene methylation patterns, and other characteristics were also consistent with those of embryonic stem cells. The iPS cells were able to differentiate into heart muscle cells, as well as other cell types.

Conclusion The four genes induced differentiated skin cells to become pluripotent stem cells, with characteristics of embryonic stem cells.

Medical Applications of Biotechnology

- The identification of human genes whose mutation plays a role in genetic diseases.
- The identification of genes that play a role in a number of “nongenetic” diseases, from arthritis to AIDS, by influencing susceptibility to these diseases.
- Finding genes that are turned on or off in particular diseases. These genes and their products are potential targets for prevention or therapy.

Diagnosis and Treatment of Diseases

Scientists can diagnose many human genetic disorders by using PCR and primers corresponding to cloned disease genes, then sequencing the amplified product to look for the disease-causing mutation.

PCR can also be used to identify symptomless carriers of potentially harmful recessive alleles as part of genetic counseling.

Individuals can be tested by PCR and sequencing for a SNP (single nucleotide polymorphism) that is correlated with the abnormal allele. The presence of particular SNPs is correlated with increased risk for particular adverse health conditions such as heart disease, Alzheimer's disease, and some types of cancer.

Personalized Medicine

Many researchers envision a future of personalized medicine, a type of medical care in which each person's specific genetic profile can provide information about diseases or conditions for which the person is especially at risk and help make health-care decisions.

An individual's genomic information can be used to predict the benefits and risks of particular medications, an approach called pharmacogenetics. There are over 300 medications for which the Food and Drug Administration recommends genetic testing for patients prior to their use.

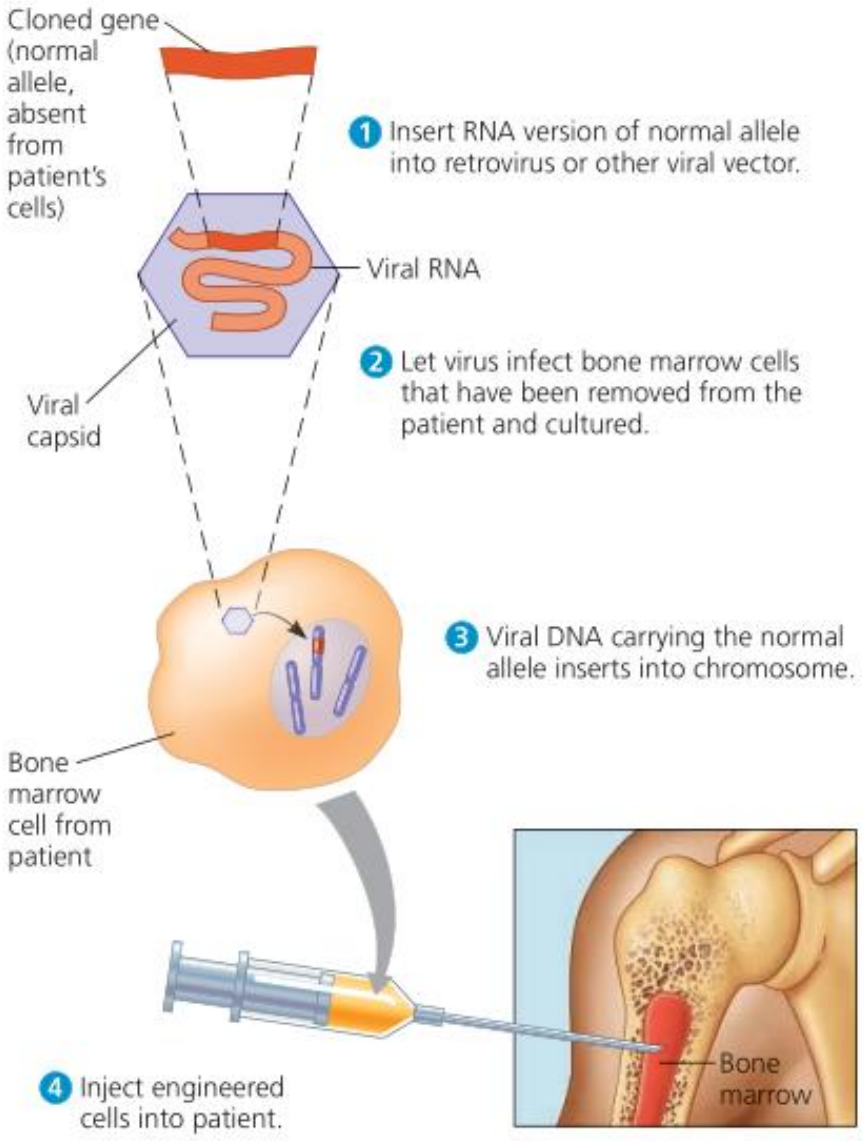
Human Gene Therapy and Gene Editing

Gene therapy—the introduction of genes into an afflicted individual for therapeutic purposes.

For gene therapy of somatic cells to be permanent, the cells that receive the normal allele must be cells that multiply throughout the patient's life. Bone marrow cells, which include the stem cells that give rise to all the cells of the blood and immune system, are prime candidates.

Gene therapy using a retroviral vector.

A retrovirus that has been rendered harmless is used as a vector in this procedure, which exploits the ability of a retrovirus to insert a DNA transcript of its RNA genome into the chromosomal DNA of its host cell (see Figure). If the foreign gene carried by the retroviral vector is expressed, the cell and its descendants will possess the gene product. Cells that reproduce throughout life, such as bone marrow cells, are ideal candidates for gene therapy.



Gene therapy using CRISPR/CAS.

In 2018, researchers reported promising results using the CRISPR-Cas9 system in an attempt to correct the genetic defect that causes sickle-cell disease.
They edited cells from patients with sickle-cell disease and injected them into bone marrow in mice.
After 19 weeks, the gene remained corrected in 20–40% of the injected cells.

Pharmaceutical Products.

The drug imatinib is a small molecule that inhibits overexpression of a specific leukemia-causing receptor. Patients in the early stages of CML who are treated with imatinib have exhibited nearly complete, sustained remission from the cancer.

Pharmaceutical products that are proteins can be synthesized on a large scale

Host cells in culture can be engineered to secrete a protein as it is made

This is useful for the production of insulin, human growth hormones, and vaccines

Protein Production by “Pharm” Animals and Plants

Transgenic animals are made by introducing genes from one species into the genome of another animal

Transgenic animals are pharmaceutical “factories,” producers of large amounts of otherwise rare substances for medical use

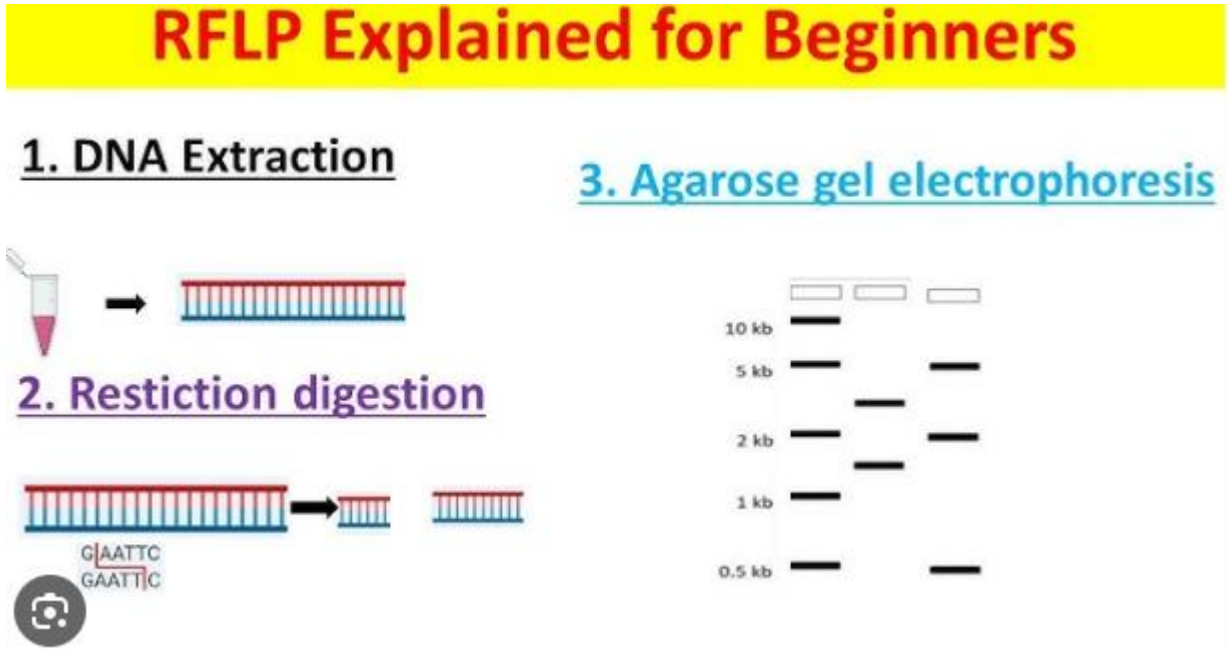
“Pharm” plants are also being developed to make human proteins for medical use

Forensic Evidence and Genetic Profiles

An individual's unique DNA sequence, or genetic profile, can be obtained by analysis of tissue or body fluids

Genetic profiles can be used to provide evidence in criminal and paternity cases and to identify human remains

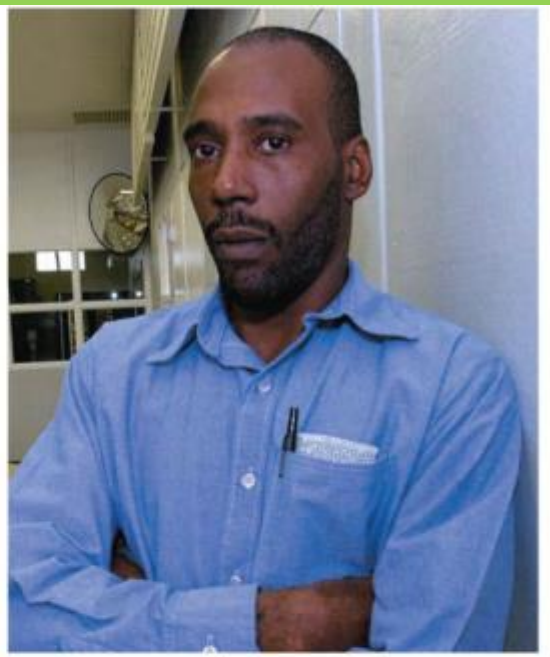
Genetic profiles can be analyzed using RFLP analysis by Southern blotting



Forensic Evidence and Genetic Profiles

- Even more sensitive is the use of genetic markers called short tandem repeats (STRs), which are variations in the number of repeats of specific DNA sequences
- PCR and gel electrophoresis are used to amplify and then identify STRs of different lengths
- The probability that two people who are not identical twins have the same STR markers is exceptionally small
- As of 2013 more than 300 innocent people have been released from prison as a result of STR analysis of old DNA evidence

(a) Earl Washington just before his release in 2001, after 17 years in prison.



Source of sample	STR marker 1	STR marker 2	STR marker 3
Semen on victim	17,19	13,16	12,12
Earl Washington	16,18	14,15	11,12
Kenneth Tinsley	17,19	13,16	12,12

(b) These and other STR data (not shown) exonerated Washington and led Tinsley to plead guilty to the murder.

Environmental Cleanup

- Many bacteria can extract heavy metals, such as copper, lead, and nickel, from their environments and incorporate the metals into compounds such as copper sulfate or lead sulfate, which are readily recoverable.
- Genetic engineering can be used to modify the metabolism of microorganisms
- Some modified microorganisms can be used to extract minerals from the environment or degrade potentially toxic waste materials
- Biofuels make use of crops such as corn, soybeans, and cassava to replace fossil fuels

Agricultural Applications

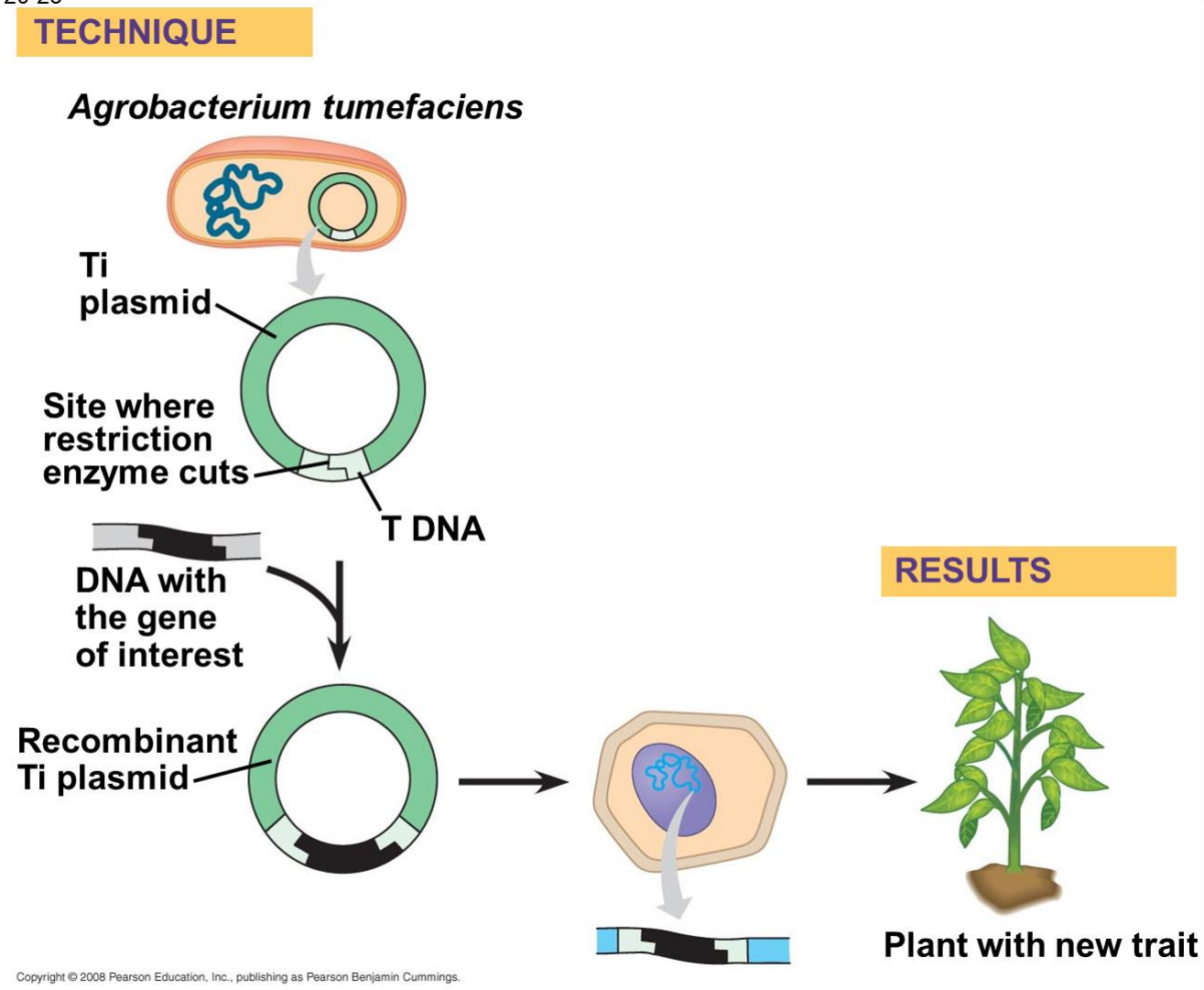
- DNA technology is being used to improve agricultural productivity and food quality
- Genetic engineering of transgenic animals speeds up the selective breeding process
- Beneficial genes can be transferred between varieties or species

Genetic Engineering in Plants

- Agricultural scientists have endowed a number of crop plants with genes for desirable traits
- The Ti plasmid is the most commonly used vector for introducing new genes into plant cells
- Genetic engineering in plants has been used to transfer many useful genes including those for herbicide resistance, increased resistance to pests, increased resistance to salinity, and improved nutritional value of crops

Genetic Engineering in Plants

Fig. 20-25



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Safety and Ethical Questions Raised by DNA Technology

Potential benefits of genetic engineering must be weighed against potential hazards of creating harmful products or procedures

Most public concern about possible hazards centers on genetically modified (GM) organisms used as food

Some are concerned about the creation of “super weeds” from the transfer of genes from GM crops to their wild relatives